Forecasting Petty Crimes During Major Holidays in the United States

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Additional contributor's names have been omitted for their privacy.

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Problem Statement

Holidays present unique challenges for police departments in preparing duty rosters and personnel assignments. Law enforcement agencies must decide the appropriate number of officers on duty during times when individuals prefer to be at home (Cohn and Rotton, 2003). It is commonly assumed that crime rates, particularly petty crimes, increase on certain holidays, such as New Year's Eve, due to factors such as, increased alcohol consumption and a higher probability of aggressive behavior. However, empirical research on holiday crime patterns remains limited, with many existing studies being outdated or inconclusive.

Early research provides mixed findings on crime trends during holidays. Lester (1979) found that homicides in the U.S. occurred disproportionately on major holidays. Similarly, Templer, Brooner, and Corgiat (1983) reported an increase in police service calls on national and local holidays. Conversely, other studies, such as Rotton and Frey (1985), failed to establish a consistent pattern indicating higher crime rates on holidays.

Several avenues have sought to spread awareness of crime spoiling the holiday spirit: Lloyd (2022) shared stories of victims that suffered car and workplace break-ins as well as an instance of a volunteer organization robbed while attempting to bring the Christmas spirit to the community by decorating houses for free, Kaufman (knklawfirm.com) cited desperation and opportunity as reasons for increased levels of crime, while Justice (2015/17) and nnw.org provided tips to lower risk and resources for victims.

Given the limited and conflicting evidence, this study aims to forecast the occurrence of petty crimes during major holidays in the United States. By leveraging historical crime data and predictive modeling techniques, this project seeks to provide insights that can assist law enforcement agencies in resource allocation and crime prevention strategies during holiday periods.

Data Source

Petty crimes are defined as infractions less serious than felonies or misdemeanors, such as pick pocketing, bag snatching, and street crime. For our project, we will be considering shoplifting, burglary, and pocket-picking, all of which involve taking coveted items. Our data covers 2020-2024, but some data sources only have data up to 2023, with this blocker in mind, we intend to train our data up to 2022, forecast 2023, and compare to the actual 2023 numbers. We also understand that data may not be available at every aggregate level, i.e. daily or weekly, monthly, so we will be addressing those pitfalls as we work through this project.

The data comes from the Federal Bureau of Investigation (FBI) Crime database, which pulls from the Uniform Crime Reporting (UCR) Program. The UCR Program has been generating reliable statistics for use in law enforcement since 1930. The data includes more than 18k city, university and college, county, state, tribal, and federal law enforcement agencies; participation is voluntary, so data submission(s) may be incomplete or absent in some years. The latest data releases are found in the FBI's Crime Data Explorer, while historical publications, up to 2019, are housed on the UCR publications page.

We will be collecting data at the national and state level on burglary, shoplifting, and pocket-picking. This data is collected on a month-by-month basis, and can be downloaded in 10-year, 5-year, and 2-year increments. To make sure our training data is robust, we will work with a 10-year dataset. Below is a sample of the data in its raw format:

series	02-2015	03-2015	04-2015	05-2015	06-2015	07-2015	08-2015
United States	7342	9945	10394	11031	11462	12050	11678
United States Clearances	4623	6097	6336	6869	7177	7408	7238

The datasets we will be using for more minutiae details will be provided by city police departments, such as Los Angeles and Boston, which are collected from data.gov. These cities were chosen because their police departments update on a daily, weekly, or monthly basis and they represent major cities.

Below is a sample of the data in its raw form:

Date Rptd	DATE OCC	AREA NAME	Crm Cd Desc	Premis Desc
3/1/20 0:00	3/1/20 0:00	Wilshire	VEHICLE - STOLEN	STREET
2/9/20 0:00	2/8/20 0:00	Central	BURGLARY FROM VEHICLE	BUS STOP/LAYOVER (ALSO QUERY 124)
11/11/20 0:00	11/4/20 0:00	Southwest	BIKE - STOLEN	MULTI-UNIT DWELLING (APARTMENT, DUPLEX, ETC)

These datasets will be useful in comparing the results we have on the monthly level, in addition to seeing if major cities have a higher rate of petty crime during holiday seasons.

Methodology

ARIMA

Statistical models play a significant role in this analysis, and one effective approach is the use of time-series anomaly detection models. This study employs the Autoregressive Integrated Moving Average (ARIMA) model for anomaly detection, given that holidays are specific points in time, making point anomaly detection particularly relevant. A point anomaly occurs when a single data instance significantly deviates from the expected pattern. The ARIMA model characterizes petty crime behavior by learning temporal dependencies and forecasting expected crime levels. The model consists of three parameters: the autoregressive parameter (p), the number of differencing steps (d), and the moving average parameter (q) (Pena, Assis, and Proença, 2013).
The ARIMA model can be written as $ARIMA(p;d;q)$.

The general ARIMA equation is given by:

zt=i=∑i=1pØizt−i=εt−∑j=1qθjzt−j

SARIMA

We also used the Seasonal Autoregressive Integrated Moving Average (SARIMA) model for petty crime predictions and anomaly detection. The SARIMA model is a time series model like ARIMA, but with an extension to account for seasonal patterns (Wu et al., 2025). The SARIMA model is also written as $SARIMA(p,d,q)(P,D,Q)[s]$. Where p is the autoregressive order, d is the difference order, and q is the moving average order (Wu et al., 2025). $P, D, and\ Q$ determines the seasonal autoregressive order, seasonal difference order, and seasonal moving average order, respectively.

Before analyzing any time series data, we need to ensure the data is stationary. Without stationarity, interpreting the results becomes problematic (Manuca and Savit, 1996). In general, stationarity is required for valid forecasts and future predictions in time series analysis (Petrică, Stancu, & Ghițulescu, 2017).

To test for stationarity, we used the Augmented Dickey-Fuller (ADF) unit root test. If the data is not stationary, we applied transformations, such as differencing, logarithmic transformation, or both, depending on the test results. Differencing is about calculating the value that involves the difference between today's value and yesterday's value.

XGBoost

XGBoost is a specific implementation of the gradient boosting algorithm, which consists of three primary steps: defining an appropriate loss function, constructing a weak learner to generate predictions, and sequentially building an additive model to refine those predictions. A key advantage of gradient boosting lies in its flexibility. Once a suitable loss function is selected, it can be seamlessly integrated into the boosting framework (Ramraj, Uzir, Sunil, & Banerjee, 2016).

Typically, decision trees are employed as weak learners in gradient boosting. These trees are constructed in a greedy fashion, where each subsequent tree aims to minimize the residuals from

the previous model. To preserve the nature of weak learners, regularization constraints are often applied, ensuring that each tree remains simple while still improving the overall prediction accuracy.

Although XGBoost follows the general principles of gradient boosting, it introduces several enhancements, including parallelized tree construction (i.e., multi-threaded implementation) and additional regularization techniques to prevent overfitting. These innovations contribute to its superior performance and computational efficiency compared to traditional gradient boosting algorithms (Ramraj, Uzir, Sunil, & Banerjee, 2016).

To supplement our SARIMA forecasting, we incorporate XGBoost, to predict petty crimes in 2023. XGBoost is capable of handling complex (i.e., non-linear) relationships in the data (Murari et al., 2024).

Anomaly detections

We applied an anomaly detection technique to test our hypothesis that petty crime rates during holiday months are higher than the average rates in non-holiday months. We selected major U.S. holiday months, including January (New Year), July (Independence Day), October (Halloween), November (Thanksgiving), and December (Christmas). To identify anomalies, we examined crime counts during holiday months that deviated significantly from the expected patterns predicted by the preferred time series models.

Anomaly detection is conducted using the predicted values of the preferred model. The predicted values include uncertainty. To detect anomalies, we compare the predicted values with the actual values. If the actual value at $t + 1$ falls outside the forecast range, we identify that data point as an anomaly (Mason et al., 2019). We calculated confidence intervals (90-95%), depending on the petty crime category. Observations falling outside these confidence intervals were classified as anomalies (Isbister, 2015).

CUSUM

Another anomaly detection method that was used was CUSUM change detection. CUSUM is commonly used as a statistical process control method to provide quick notice when data points deviate from the target values beyond the desired threshold. One notable strength of CUSUM is that it's able to detect relatively small shifts, which is useful for this project, since we don't expect increases in crime caused by holidays to last beyond the month the holiday occurred in, so we need to be able to identify the anomalies/changes as soon as they happen rather than having a lagging indicator as some other change detection methods may provide.

The CUSUM formula is:

$$S_i = \max[0, S_{i-1} + (x_i - \mu - C)]$$

Where:

- $S_0 = 0$
- x_i is the current data point
- μ is the mean value of all the data points

- C is the allowed variance level

Once the value of S_i reaches a pre-specified threshold (T), it will signify a statistical increase having taken place. A generally accepted practice is to set C = 0.5σ (σ=standard deviation) and T = 3σ, which is what will be tested for the crime data.

Box and Whisker

While CUSUM providing notice of changes for months is useful, to determine if the total number of changes for any months could be considered outliers and support our theory that holiday months have higher crime levels, we'll be using box and whisker plots (also known as boxplots). Commonly used for outlier detection, boxplots function by splitting the data into quartiles and displaying the "box" with its border being the two center quartiles and the entire range of Q1-Q3 also being known as the "Inner Quartile Range" (IQR). The "whiskers" that extend in either direction calculated by Q1 – 1.5 * IQR and Q3 + 1.5 * IQR to determine the minimum and maximum values respectively. Any values that are outside of the calculated minimum/maximum range will be deemed outliers. If our theory of crime being higher during holiday months is true, then we should see enough changes in the holiday months for them to be identified as outliers beyond the maximum range.

Model Evaluation

We evaluated our model's performance using various forecasting evaluation metrics. We used crime data up to 2022 as the training sample and predicted petty crime rates for 2023. The predictions were then compared to actual petty crime counts using the following error metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). The error matrixes are computed as follows (Fang & Lahdelma, 2016):

$$RMSE = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(\hat{y}_t - y_t)^2}$$

$$MAE = \frac{1}{N}\sum_{t=1}^{N}|\hat{y}_t - y_t|$$

$$MAPE = \left(\frac{1}{N}\sum_{t=1}^{N}\left(\frac{|\hat{y}_t - y_t|}{y_t}\right)\right)$$

Where $\hat{y}_t$ is the predicted petty crime count at time t, and y_t is the observed petty crime count at time t. N is the total number of predictions.

Results

National Level Shoplifting Data

Figure A.1 in the appendix shows the monthly number of shoplifting cases in the United States from 2015 to 2023. There is a significant upward trend in the number of shoplifting cases during this period. To assess stationarity, we conducted an Augmented Dickey-Fuller (ADF) test, which indicated that the raw data was non-stationary. To address this, we applied a log transformation followed by first differencing. After these transformations, the ADF test confirmed that the data had become stationary. (Figure A.2 in the appendix presents the transformed, stationary data).

Panels 1 and 2 of Figure A.3 in the appendix display the estimated autocorrelation function (ACF) and partial autocorrelation function (PACF) plots of the transformed series. The ACF and PACF plots show a strong correlation at lag 1, while the remaining lags exhibit mostly small autocorrelations. We did not observe any significant decay patterns or negative spikes, except for a notable negative spike at lag 10.

After analyzing the ACF and PACF plots, we performed parameter tuning by testing different p, d, and q values. We found that the SARIMA $(1,1,1)(0,0,0)_{12}$ model had the lowest Akaike Information Criterion (AIC) value, confirming that it was the most suitable model for our predictions. Using this final model, we predicted U.S. shoplifting numbers for 2023 and compared them with the actual observed shoplifting numbers for the same period. Figure 1 visualizes these values.

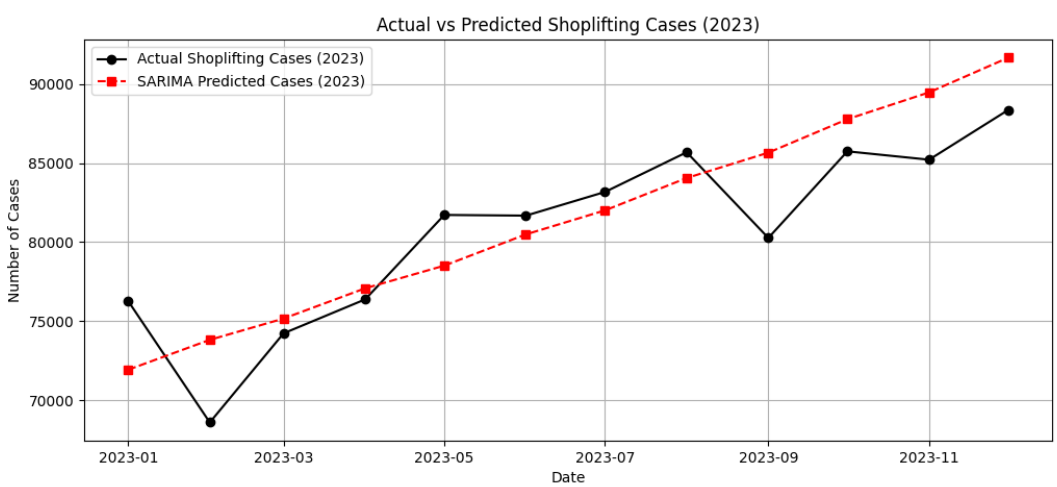


Figure 1: US shoplifting forecasts for 2023 vs actual values

Anomaly detection in shoplifting cases

After making predictions for 2023, we tested whether shoplifting numbers were significantly higher during holiday months. Using 90% confidence intervals, we evaluated whether shoplifting cases during holiday months were significantly higher than non-holiday months. (Figure A.4 presents the anomaly detection plot for nationwide shoplifting). After conducting the confidence interval analysis, we quantified the effects using z-scores. Table 1 reports the z-scores for specific holiday months, comparing the predicted and observed shoplifting cases. Our findings indicate that there were no severe anomalies during the 2023 holiday months, suggesting that crime counts remained within the expected range. Most z-scores were negative, indicating that shoplifting was lower than expected in these months. However, these negative z-scores were not statistically significant. Additionally, no extreme crime spikes were observed during the holiday months.

Table 1: Anomaly detection for national level shoplifting using z-scores

Holiday Month	Actual Cases	Forecasted Cases	Residual	Z-scores
July 2023	83170	82005	1165.05	0.65
October 2023	85738	87770	-2031.79	-0.38
November 2023	85209	89465	-4256.08	-1.09
December 2023	88349	91655	-3306.14	-0.78

Note: January was not flagged as an anomaly 2023

National level burglary data

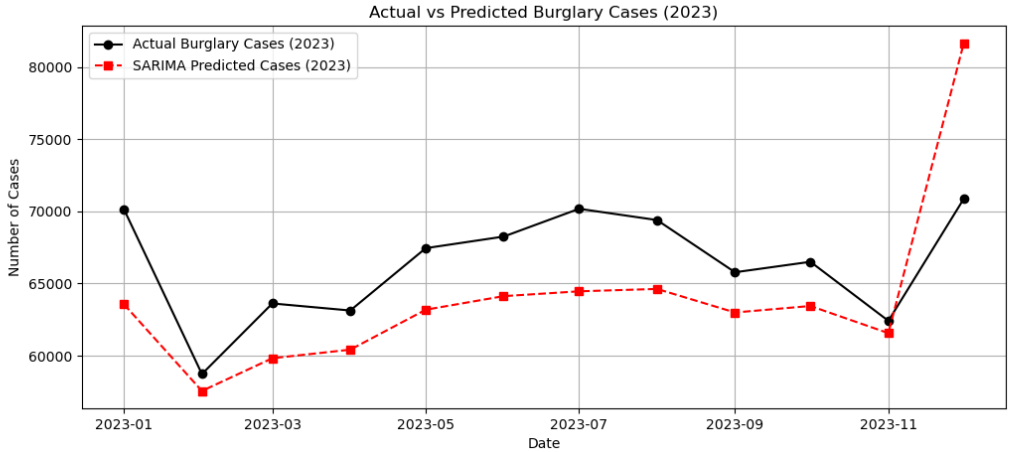


Figure 2: US burglary forecasts for 2023 vs actual values

After examining the ACF and PACF plots (Figure A.7), and some model selection using parameter tuning and AIC values we set our final SARIMA model as (2,0,2) (1,0,1). [1] In this model specification, we accounted for annual seasonality.

Our best SARIMA model underpredicts burglary cases for early 2023 (Figure 2). This suggests that the model might be too conservative in capturing crime spikes, possibly due to underestimated seasonal effects. Even though, the model assumes strong seasonality, actual data deviates from that assumption for 2023. This might be because of new policies and law enforcement actions. However, a MAPE of 5.87% suggests that overall, the model performed well for the given data.

Anomaly detection in burglary cases

Table 2: Anomaly detection for national level burglary using z-scores

Holiday Month	Actual Cases	Forecasted Cases	Residual	Z-scores
July 2023	70182	64828	5354.41	0.74
October 2023	66507	63771	2736.08	0.14
November 2023	62414	61896	517.98	-0.38
December 2023	70877	82018	-11141.11	-3.08

The forecasted values (Table 2) follow the general pattern of actual burglary cases. The confidence interval suggests an uncertainty range of predictions. The z-scores suggest that there was no anomaly detection for the July, October, and December. The SARIMA model performed well in October and November months capturing seasonality effectively. However, for December 2023, the model expects a large surge in burglary cases, but actual cases were much lower.

National level pocket-picking data

For our next petty crime forecasting analysis, we selected national-level pocket-picking data from 2015 to 2024. The data exhibited an upward trend over this period. The initial ADF test did not find evidence that the data series was stationary. Therefore, we applied a transformation. After log-transforming and first differencing the data, the ADF test concluded that the series was stationary.

The ACF plot shows (Figure A.7) a sharp drop after lag 0, with all subsequent lags remaining within the confidence interval. This suggests that the time series does not exhibit significant

[1] For both national and city-level petty crime datasets, we initially assessed the stationarity of the time series using the ADF test. Based on the test results, we applied appropriate data transformations—such as first-order differencing, second-order differencing, or logarithmic differencing—to achieve stationarity. To explore underlying data patterns and temporal dependencies, we examined the ACF and PACF plots. Subsequently, we conducted model selection through iterative testing and comparison of candidate models using the Akaike Information Criterion (AIC), selecting the model that minimized the AIC value as the most appropriate specification for each dataset.

autocorrelation beyond the first lag. Similarly, the PACF plot follows the same pattern. After tuning model parameters and analyzing autocorrelation plots, we selected SARIMA (2,1,1) (0,0,0) as our best model.

Figure 3 shows the model forecast for 2023 national pocket-picking cases compared to the actual number of reported cases. Our model successfully captures the upward trend in pocket-picking cases throughout 2023. However, it underestimates peak crime months**.**

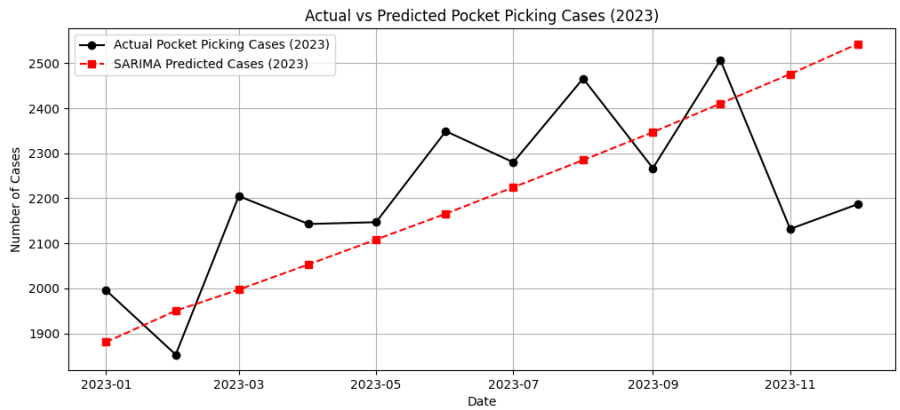


Figure 3: US pocket-picking forecasts for 2023 vs actual pocket-picking cases

Anomaly detection in pocket picking cases

Table 3: Anomaly detection using z-score

	Actual Cases	Forecasted Cases	Residual	Z-scores
July 2023	2224	2280	55.47	0.26
October 2023	2410	2507	96.50	0.48
November 2023	2476	2132	-343.87	-1.91
December 2023	2543	2187	-356.02	-1.98

Table 3 presents the Z-scores based on the residuals. The months of July and October have small residuals and low Z-scores, indicating that these months fall within the expected range. Thus, no significant anomaly is observed (i.e., no substantial increase in pocket-picking cases). However, November and December exhibit large negative residuals, suggesting that these months could be considered potential anomalies.

Model evaluations using error matrices

Table 4: Error matrices for 2023 predictions

Error matrix	Shoplifting	Burglary	Pocket-picking
RMSE	3234.06	4819.33	183.83
MAE	2781.50	4001.19	153.70

MAPE	3.50%	5.87%	6.98%

Table 4 presents the different model evaluations using error matrices. For national shoplifting data, the SARIMA model performs relatively well, with a low MAPE of 3.50%**.** Among the three crime types, burglary predictions have the highest error rates**.** The MAPE of 5.87% suggests a moderate level of accuracy, which may be due to the higher variance in burglary cases, making them more difficult to predict accurately.

For pocket-picking predictions, the model has the smallest RMSE and MAE values compared to the other crime types. This is because of the lower overall volume of national pocket-picking cases. However, the MAPE is the highest**,** indicating that the model has the largest percentage error relative to shoplifting and burglary cases.

XGBoost method for forecasting

XGBoost forecasting results

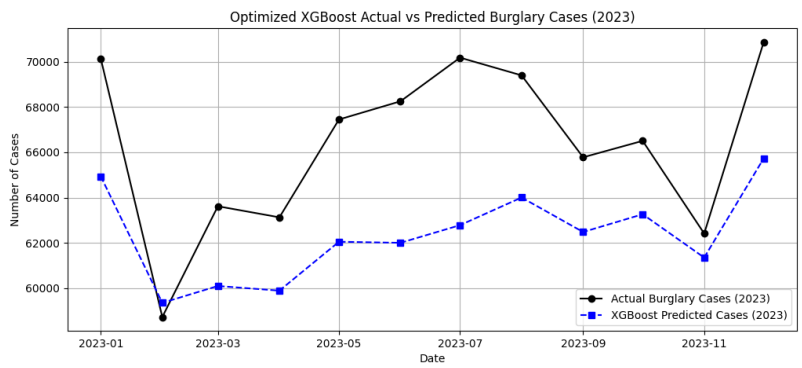


Figure 4: US burglary forecasting for 2023 using XGBoost vs actual burglary cases

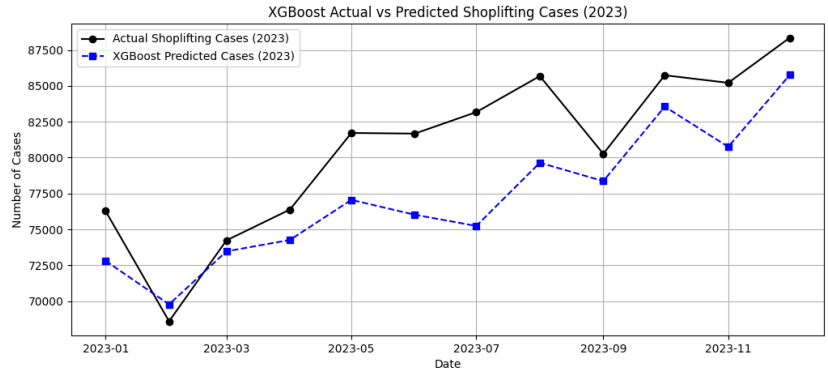


Figure 5: US shoplifting forecasting for 2023 using XGBoost vs actual burglary cases

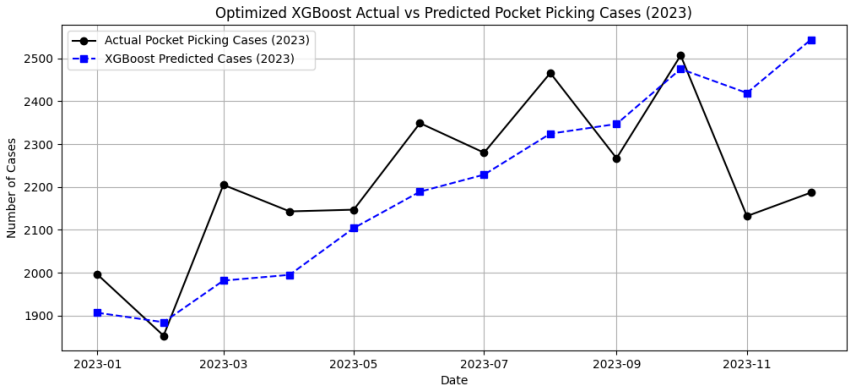


Figure 6: US pocket-picking forecasting for 2023 using XGBoost vs actual burglary cases

Model evaluation using error matrices

Table 5: Error matrices for 2023 predictions using XGBoost method

Error Matrix	Burglary data	Shoplifting data	Pocket-picking data
RMSE	4576	4149	170
MAE	4148	3580	137
MAPE	6.12%	4.36%	6.22%

In our national petty crime data, we observe that some of the XGBoost models (Figure 4-6) did not perform as well as the SARIMA models (Table 5). This may be due to the strong seasonality effect, which SARIMA captures better than the XGBoost algorithm. XGBoost does not inherently model seasonality and relies on lag features, which might not fully capture long-run cycles (Sen et al., 2024). Additionally, if the data exhibits mostly linear patterns, SARIMA models tend to perform better. When the underlying pattern is linear with seasonal effects, SARIMA models may yield more accurate results, as XGBoost treats time series as a regression problem and might struggle with long-term trend shifts.

City-level data – Los Angeles (LA)

We explored city-level petty crime data to examine whether it follows the same pattern as the national-level data and to determine if petty crime levels during holiday months are higher than non-holiday months. For the city-level data, we had additional variables such as area name,

premise description, weapon description, and location. However, the data was inconsistent, and some missing values were observed. Therefore, we performed data cleaning, and since the original dataset was at the daily level, we aggregated it to the monthly level to ensure consistency with the national-level analysis.

Given the additional variables, we were able to use the SARIMAX model in some cases to improve model performance. However, due to data inconsistencies and limitations, we could not apply SARIMAX to all three crime categories. For the Los Angeles city data, only burglary cases had sufficient data quality to proceed with the SARIMAX model. Like the national-level analysis, we first made the data stationary before performing time series analysis. The best models were selected based on cross-validation and AIC statistics. The reported results reflect the final selected models.

LA burglary data

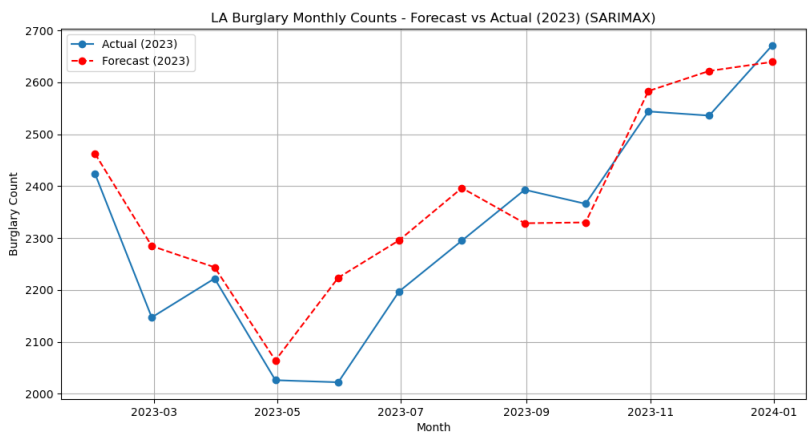


Figure 7: SARIMAX model forecasting for LA burglary data- 2023

Table 6: Anomaly detection using z-score method- LA city burglary cases 2023 holiday months

Month	Actual Burglary Count	Predicted Burglary Count	Z-Score	Anomaly
Jan. 2023	2424	2462.72	-0.2736	No
July 2023	2295	2396.46	-0.717	No
Oct. 2023	2544	2583.33	-0.278	No
Nov. 2023	2536	2622.04	-0.6081	No
Dec. 2023	2671	2639.4	0.2233	No

For the LA city burglary data, we applied data transformation by first differencing, as the ADF test indicated that the original data was not stationary. After examining the ACF and PACF (Figure A.8) plots and selecting the model based on the AIC criterion, we found that the best model for the LA city burglary data was the SARIMA (1,1,1) $(1,1,1)_{12}$ model. Additionally, the burglary type was used as the exogenous variable, while all other variables were dropped due to inconsistencies and missing values.

Figure 7 presents the SARIMAX forecast for burglary cases in 2023. We then used the forecasted values for the holiday months to detect anomalies using z-scores. However, we did not find any significant anomalies in burglary cases in Los Angeles for 2023 (Table 6).

LA shoplifting data

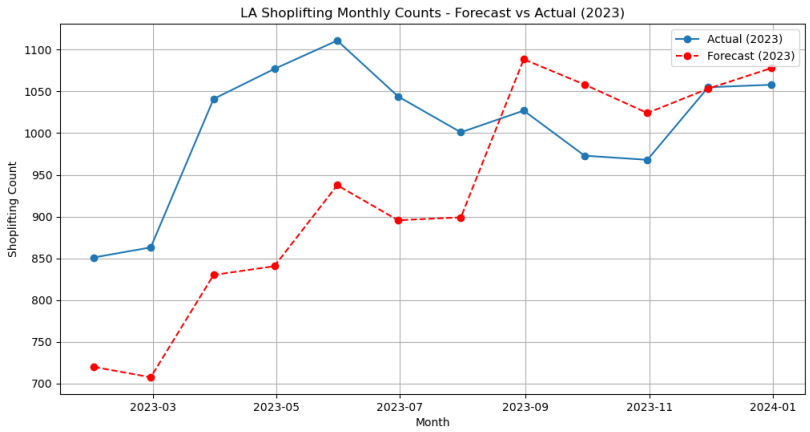


Figure 8: SARIMA model forecasting for LA city shoplifting cases- 2023

Table 7: Anomaly detection using z-score method for LA city shoplifting cases – 2023 holiday months

Month	Actual Shoplifting Count	Predicted Shoplifting Count	Z-Score	Anomaly detection
Jan. 2023	851	719.68	1.5499	No
July 2023	1001	899.04	1.2033	No
Oct. 2023	968	1024.12	-0.6623	No
Nov. 2023	1055	1053.37	0.0192	No
Dec. 2023	1058	1077.84	-0.2342	No

For the LA shoplifting data, we followed the same data processing steps, including first differencing. After obtaining stationary data, we plotted the ACF and PACF graphs (Figure A.9). Based on model selection, we found that the best model for predicting LA shoplifting cases was SARIMA (2,1,1) $(1,1,1)_{12}$. The SARIMA forecast compared to the actual cases is presented in Figure 8.

We then used the preferred model to forecast shoplifting cases for 2023 and analyzed whether any anomalies were observed during the holiday months (Table 7). No significant anomalies were detected in the LA city shoplifting cases for 2023.

LA pocket-picking data

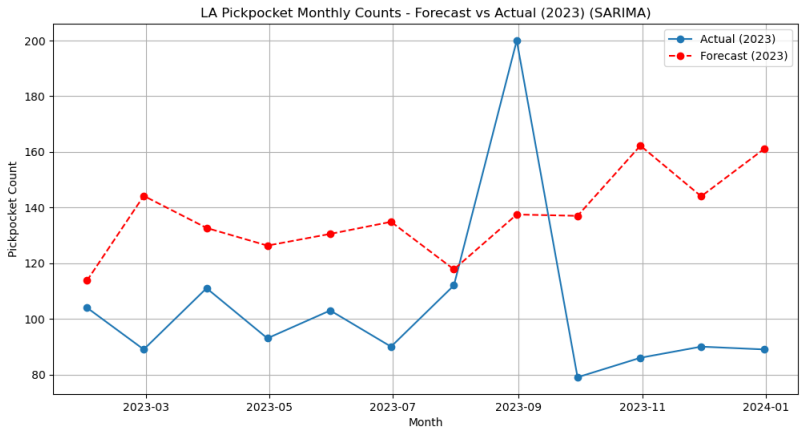


Figure 9: SARIMA model forecasting for LA city pocket-picking cases- 2023

Table 8: Anomaly detection using z-score method for LA city shoplifting cases – 2023 holiday months

Month	**Actual Pickpocket Count**	**Predicted Pickpocket Count**	**Z-Score**	**Anomaly detection**
Jan 2023	104	113.72	-0.8655	No
July 2023	112	117.78	-0.5141	No
Oct 2023	86	162.23	-6.786	Yes
Nov. 2023	90	144.02	-4.8088	Yes
Dec. 2023	89	161.03	-6.4115	Yes

For the LA city pocket-picking data, we found that the SARIMA $(1,1,1)(1,1,1)_{12}$ model was the preferred model. After selecting the best model, we generated predictions for 2023. Figure 9 presents the SARIMA forecast compared to the actual pocket-picking cases.

We observed a significant peak in actual pocket-picking cases in September 2023; however, the SARIMA model was not able to capture this variation. As with the other cases, we performed anomaly detection (Table 8). Although some anomalies were detected, they represented unexpectedly low pocket-picking activity rather than an increase. Model performance was poor, with a MAPE of 45.22% (Table 9). Therefore, we conclude that no significant anomalies were detected.

Table 9: Error matrices for 2023 LA petty crime predictions

Error Matrix	Burglary data	Shoplifting data	Pocket-picking data
RMSE	91	135	43
MAE	75	155	43
MAPE	3.34%	11.52%	45.22%

Comparing the model performances for the LA petty crime data, all three models performed worse than the national petty crime forecasting models (Table 9). The shorter time and data inconsistencies, including missing values, are likely the main factors contributing to the poor model predictions for the LA data. Additionally, the pocket-picking model reported the worst performance, with a MAPE of 45.22%, compared to the burglary and shoplifting predictions.

City-level data – Boston

For the Boston city-level data, we had additional variables that could be used as exogenous variables for SARIMA, such as, the district, offense description, street, and latitude/longitude. Before performing modelling, a simple exploratory data analysis (EDA) was conducted to get a better feel for the data, so each category: pickpocketing, shoplifting, and burglary, had a line chart demonstrating crime over time, in years, and a crime count to note any major anomalies. Afterwards, each dataset was subjected to an ADF test, to determine whether the data was stationary or non-stationary. If it was determined to be non-stationary, based on a p-value greater than 0.05, then differencing would be applied. Since pickpocketing and burglary were found to be stationary ($p < 0.05$), no differencing was applied, which simplifies the modeling process and preserves interpretability Despite following the same SARIMA modeling procedure as used on the LA data, the Boston datasets for burglary, pickpocketing, and shoplifting produced only a single forecast value. This suggests a possible issue with either model convergence, parameter specification, or the exogenous variable integration pipeline. Further debugging or alternative modeling approaches (e.g., Prophet, XGBoost with time features) may be needed to address this limitation. Despite this hiccup, the Boston data yielded similar insights as the LA data.

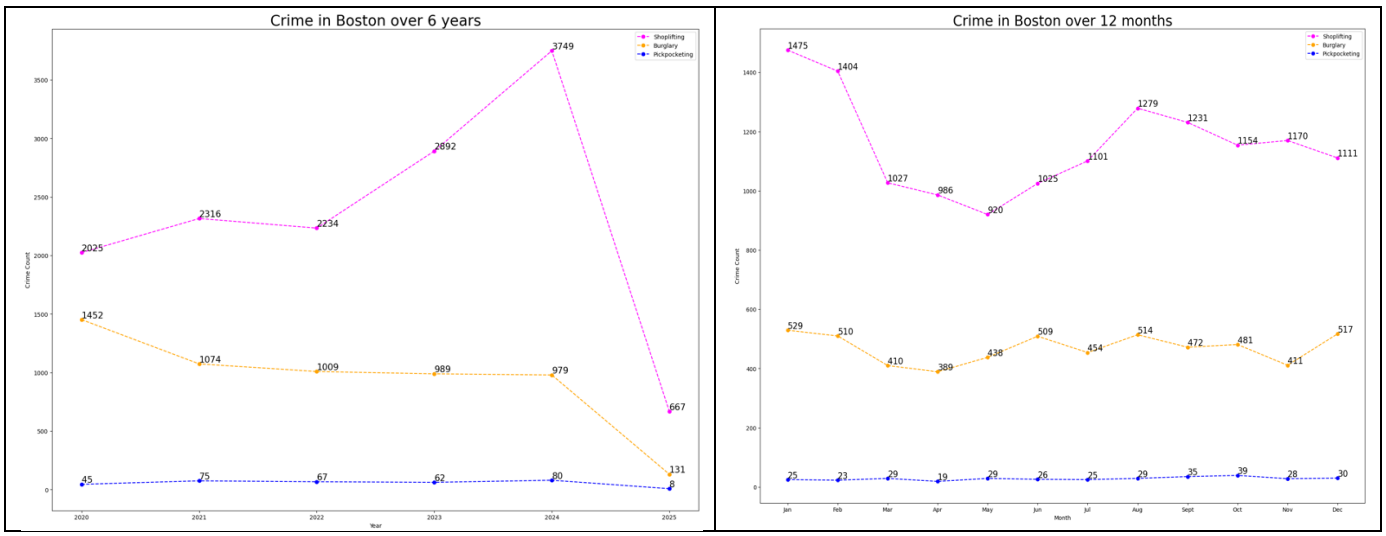


Figure 10: Crime in Boston over the years, and months, respectively

Boston Burglary

For Boston burglary, one-hot encoding, which is the process of turning categorical values into numerical values, was applied to the district, to see if that made a difference in the model's predictive powers. CUSUM was also applied to determine if there was any change detected. The results of the ARIMA test can be found below. Sigma represents the variance of the residuals, which reflects noise in the data. MA.L1 and MA.L2 mean Significant moving average terms (p-values < 0.000) indicate that the model captures short-term correlations effectively. For the SARIMA test results, some nuances should be noted. For example, since RMSE is larger than MAE, it indicates the presence of some larger errors. If the RMSE is significantly higher than the MAE, it may suggest that the model has difficulty predicting certain cases (outliers or extreme values).

Table 10: ARIMA and SARIMA Results for Boston Burglary

Burglary ARIMA Results		Burglary SARIMA Results	
AIC	7861.858	**MAE**	1.4358
BIC	7878.245	**RMSE**	1.740
Sigma	5.3239	**AIC**	3671.814
MA.L1	-0.7904	**BIC**	3759.219
MA.L2	-0.1987	**Sigma**	0.4797
Skew	8.94	**Skew**	1.94
Kurtosis	202.05	**Kurtosis**	8.60

Comparing ARIMA and SARIMA for Boston burglary reveals a stark contrast. SARIMA dramatically reduces residual variance ($\sigma = 0.4797$ vs. ARIMA results of 5.3239), and achieves far lower AIC/BIC scores, indicating a substantially better fit. This reinforces the value of incorporating seasonal components. However, the elevated RMSE over MAE for the SARIMA results suggests some susceptibility to outliers remains.

Boston Pickpocketing

Table 11: ARIMA and SARIMA Results for Boston Pickpocketing

Pickpocketing ARIMA Results		Pickpocketing SARIMA Results	
AIC	388.375	**MAE**	0.2874
BIC	399.425	**RMSE**	0.4853
Sigma	0.2118	**AIC**	381.117
MA.L1	-1.0057	**BIC**	388.491
MA.L2	0.0122	**Sigma**	0.2102
Skew	4.06	**Skew**	4.10
Kurtosis	24.06	**Kurtosis**	24.38

SARIMA consistently outperforms ARIMA across AIC, BIC, and residual variance. The improvement is modest, which could indicate that seasonal patterns are weak or minimal in the data.

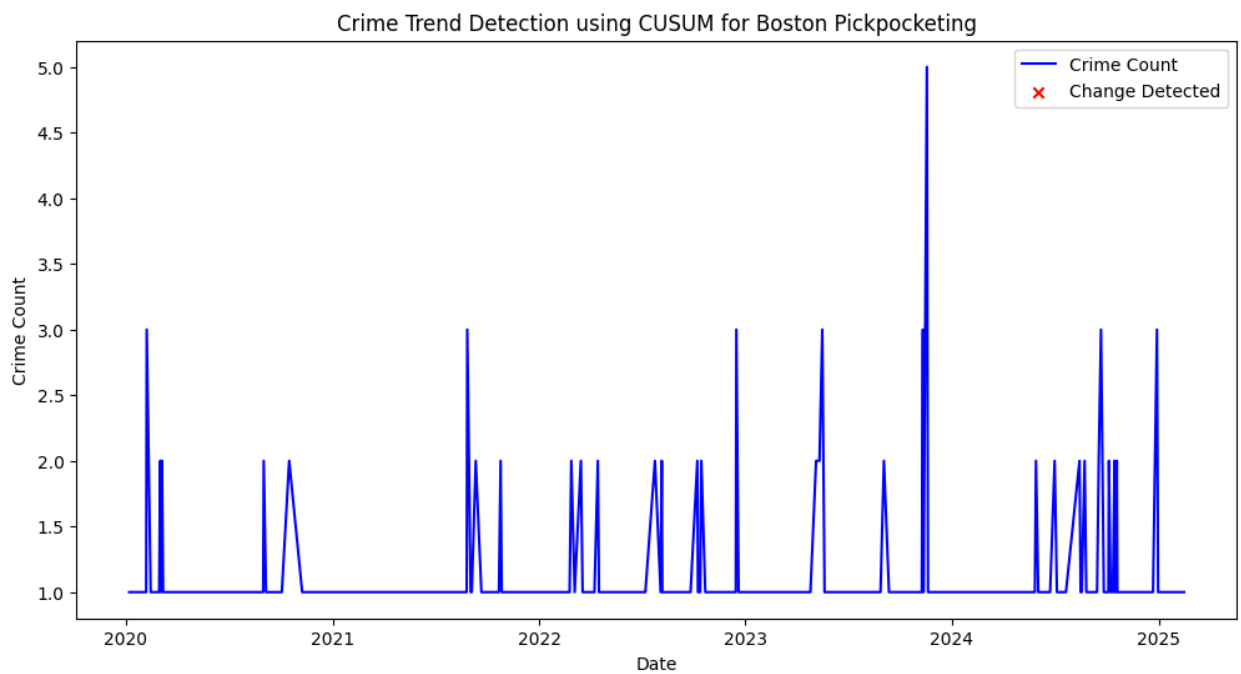


Figure 11: CUSUM for Boston Pickpocketing

Notice that there is no crime change detection for pickpocketing via CUSUM, maybe it is because there is not a lot of data to suggest any anomalies or outliers could be present to skew the data, or it is just representative of the consistent nature of this crime over time.

Boston Shoplifting

Boston Shoplifting was the most promising, out of the three categories, for achieving the desired model results. Before differencing:

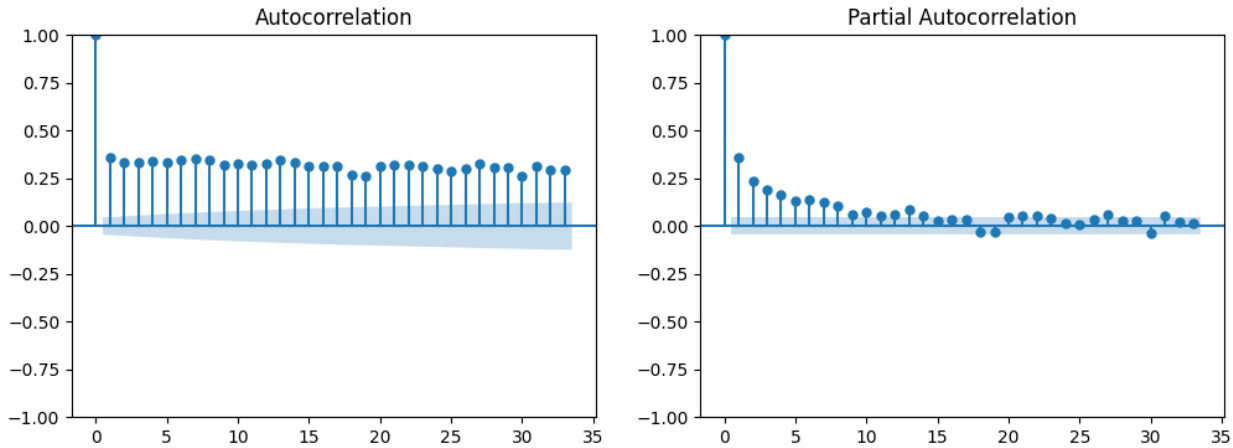


Figure 12: ACF and PACF for Boston Shoplifting before differencing is applied

After differencing:

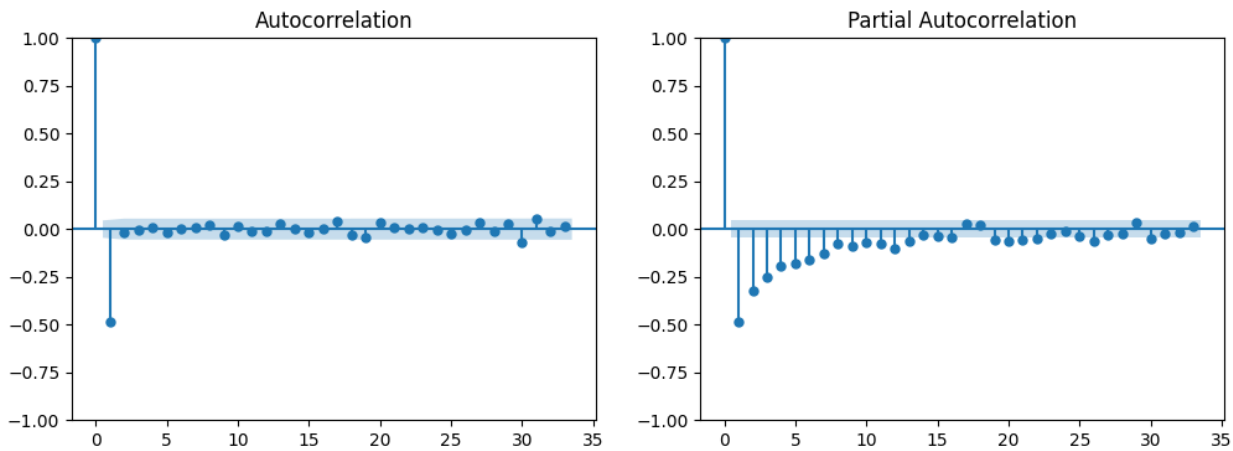


Figure 13: ACF and PACF for Boston Shoplifting after differencing is applied

Table 12: ARIMA and SARIMA Results for Boston Shoplifting

Shoplifting ARIMA Results		Shoplifting SARIMA Results	
MAE	3.0870	**MAE**	1.144
RMSE	4.046	**RMSE**	1.4615
AIC	10195.671	**AIC**	9058.704
BIC	10206.671	**BIC**	9069.704
Sigma	16.2749	**Sigma**	8.7495
Skew	-0.06	**Skew**	0.37
Kurtosis	3.13	**Kurtosis**	3.43
MA.L1	-0.997	**MAE**	1.144

Boston shoplifting data yielded the most promising SARIMA model, with balanced MAE and RMSE and low residual skew/kurtosis. This suggests that shoplifting behavior may be more stable or seasonally patterned, allowing for improved forecasting performance. As noted below, there were several instances of change detected using CUSUM, suggesting that the highs and lows of shoplifting data is consistent over time.

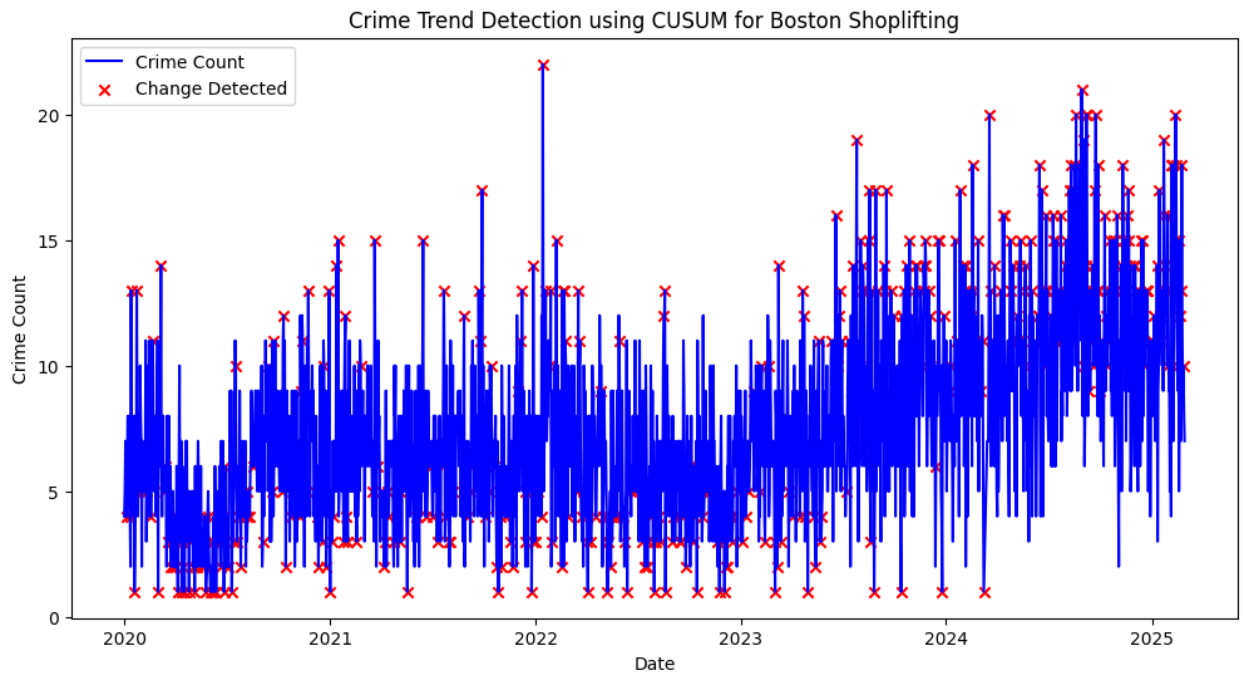
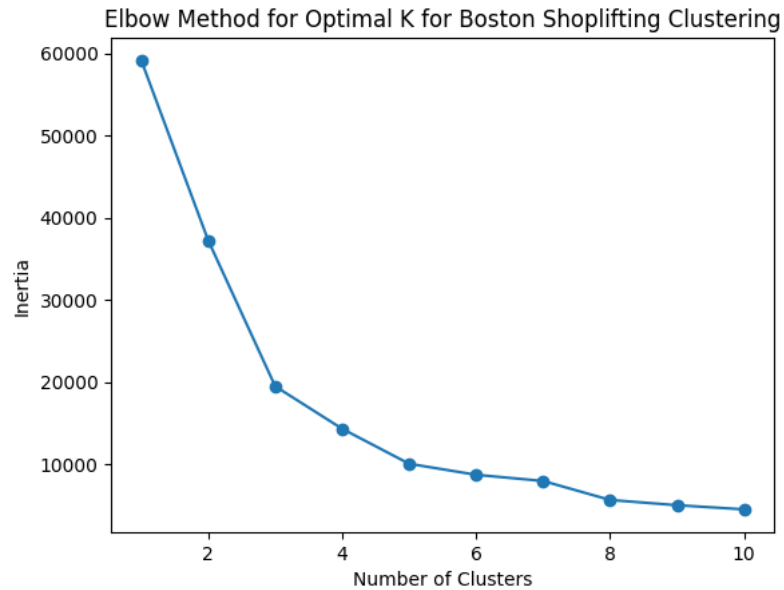
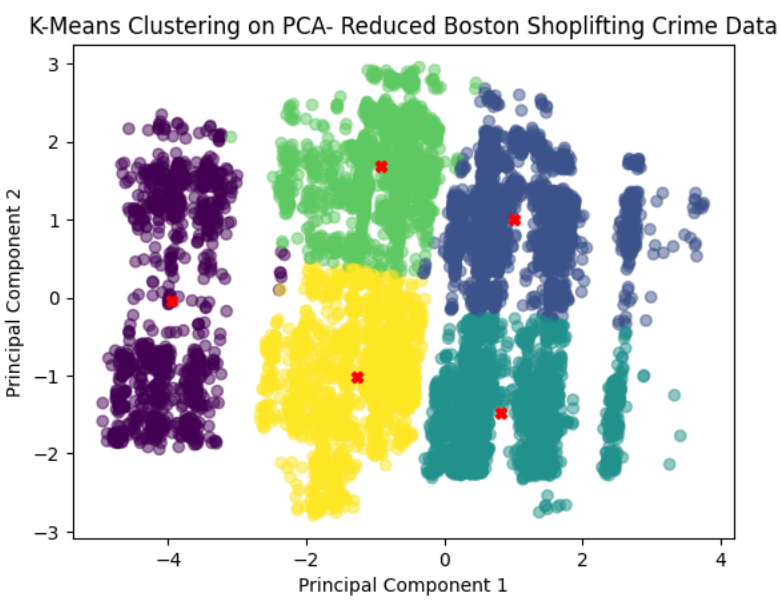


Figure 14: CUSUM for Boston Shoplifting over the years

Clustering Boston Shoplifting

A clustering method was applied to the shoplifting data because it was the most robust. The features selected were Year, Month, Hour (temporal), Latitude, and Longitude (spatial), while one-hot encoding was performed on Day of the Week and District. Selecting temporal and spatial features enhanced the analysis by adding more nuanced local patterns to be considered. After that, the data was standardized, then a Principal Component Analysis (PCA) was performed with n_components = 2, which resulted in an explained variance ratio of 0.08012909 0.0620654. The ratio suggests that PCA did not capture as much of the data's variance as hoped, indicating that some significant information was lost in the process; perhaps more components could have avoided this. Afterwards, clustering was performed using the optimal k, found after using the elbow method. The optimal k was found to be 4 or 5, while the different colored clusters in the third image represent the grouping of districts.





<- Different colors are representative of the district groupings

Figure 15: PCA and Elbow Method for Boston Shoplifitng data

CUSUM Analysis

National Level

The crime data for the US states was aggregated at the month level, so we'll be evaluating if changes are detected at a higher level in the holiday months compared to the remaining months using data from 2015-2023. For the three crime types (burglary, pocket-picking, and shoplifting), each state was evaluated while keeping track of the total instances of changes being detected for each month. Results are shown below:

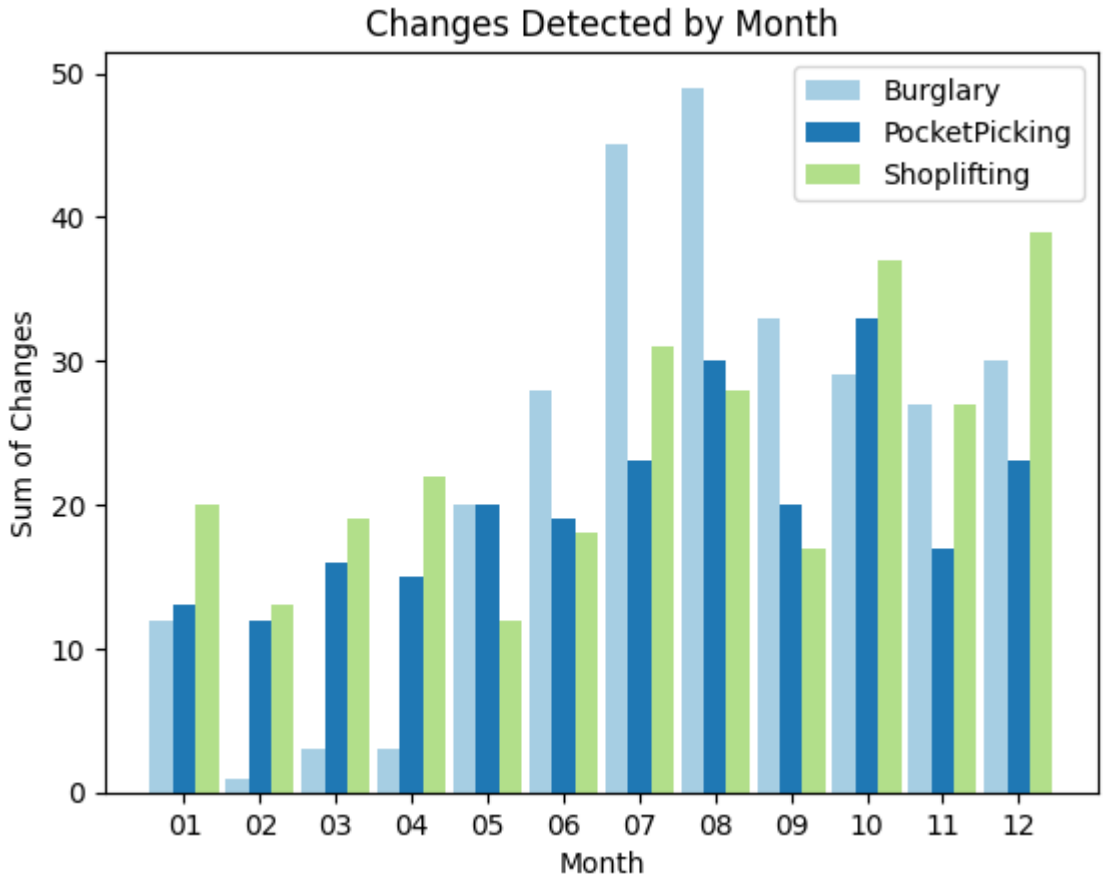


Figure 16: Total changes detected each month per crime type

Starting with a quick eye test, these results certainly show that crime levels have variance throughout the year and will increase at certain points, but there is no noticeable increase during our holiday months. Each of the crime types don't necessarily follow the same patterns, but none seem to support our hypothesis of holiday months having increased crime. For Burglary, January has very few changes overall, July has a lot, but August has even more than October to December, but still not as many as the summer months. Pocket Picking steadily increases for the most part until October, then drops off, but no month shows a significant difference. Shoplifting also trends upward over the course of the year, but once again, no single month stands out.

For a less subjective ruling, here are the boxplots for each crime type:

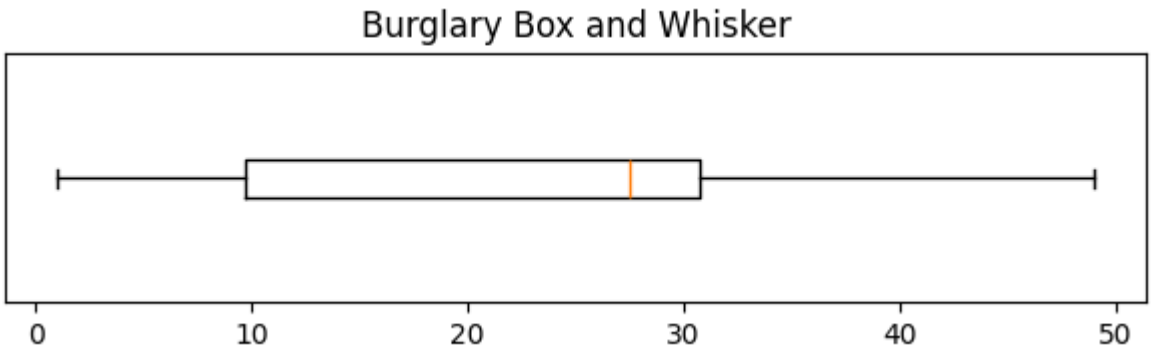


Figure 17: US burglary changes boxplot

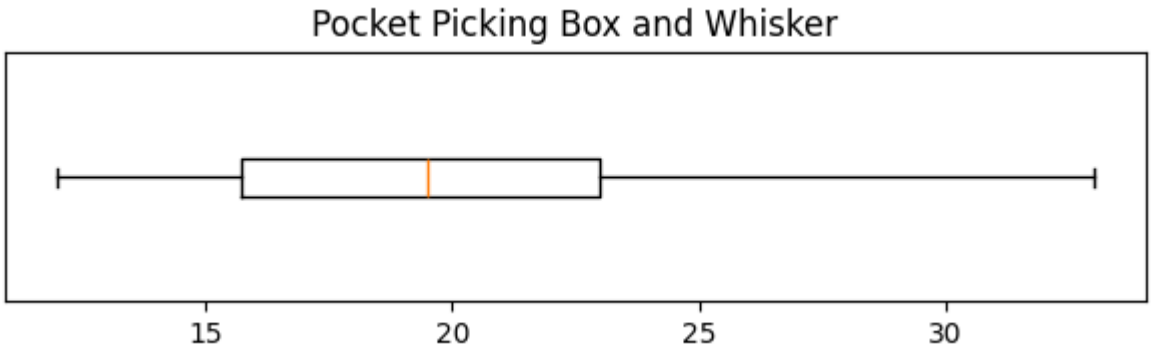


Figure 18: US shoplifting changes boxplot

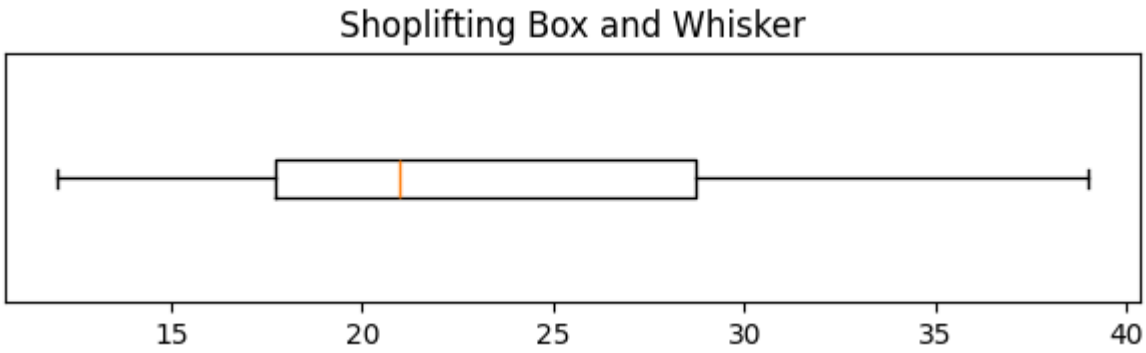


Figure 19: US pickpocket changes boxplot

Viewing these plots, we can clearly see that while the distributions for each crime type are different, there aren't any outliers. As previously stated, to consider a month to have a meaningful increase in the crime levels, we were looking for outliers, so this confirms the speculation from the eye test that the holiday months don't have statistically significant increases in crime.

State Clustering Using K-Means

After looking at the individual states, the next approach was to use K-means clustering to group the states based on their crime levels and use CUSUM to analyze each of the clusters. Using k=7, the results of the clustering are shown below, starting with burglary data:

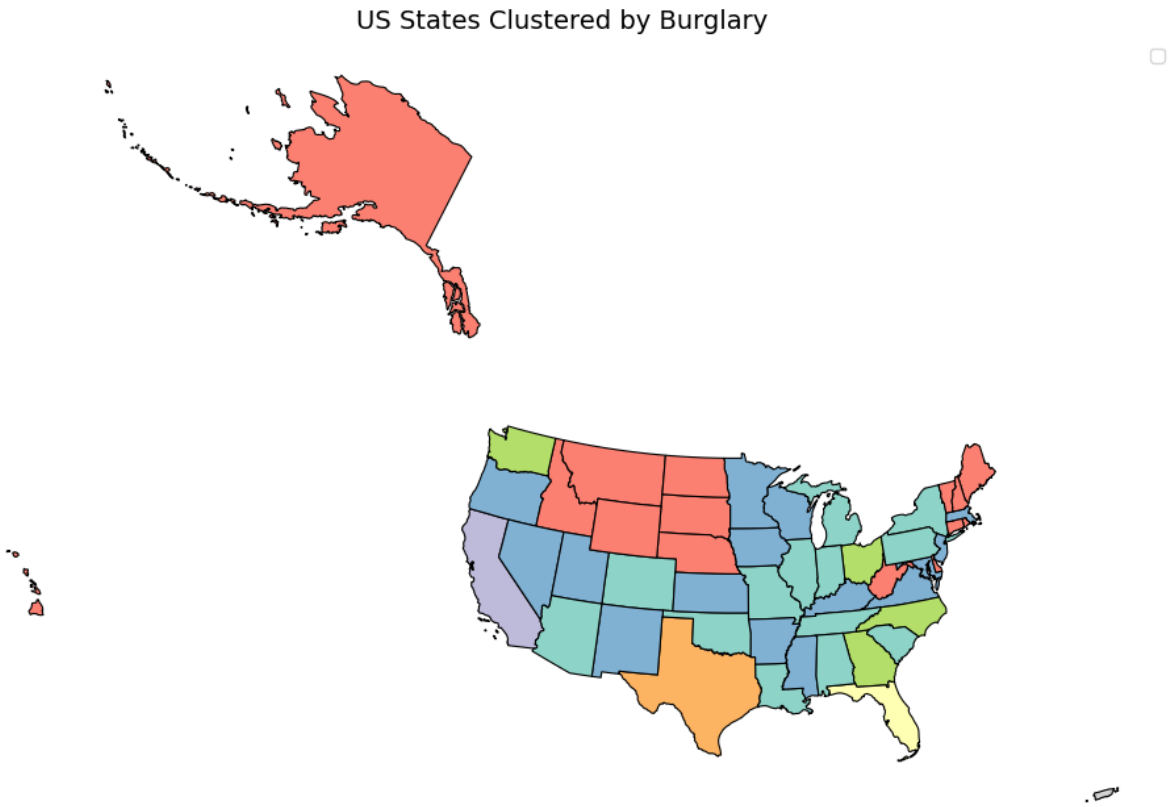


Figure 20: State clusters based on burglary data

The crime for states in each cluster was averaged and the CUSUM change detection formula was applied to those values with the results shown in Figure A.11 in the appendix. The total number of changes for each group was very low, with only one instance of a group having more than two changes in a month, this group contained Florida, which only had four instances of change in December. Overall, the holiday months don't stand out above the rest in terms of changes, and like the individual states, more changes occurred during the summer months. Below is another chart showing the aggregation of these changes, followed by the boxplot confirming that the holiday months don't show an increase in crime levels for the clustered burglary data.

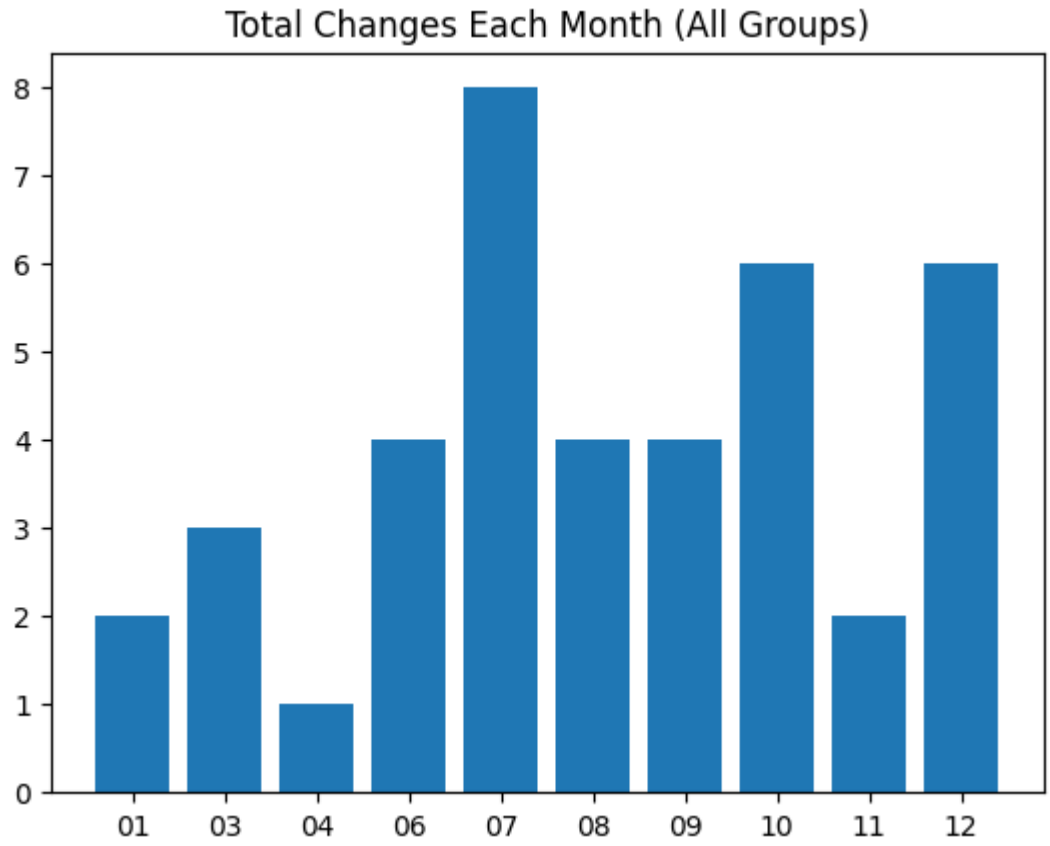


Figure 21: Clustered states burglary changes per month

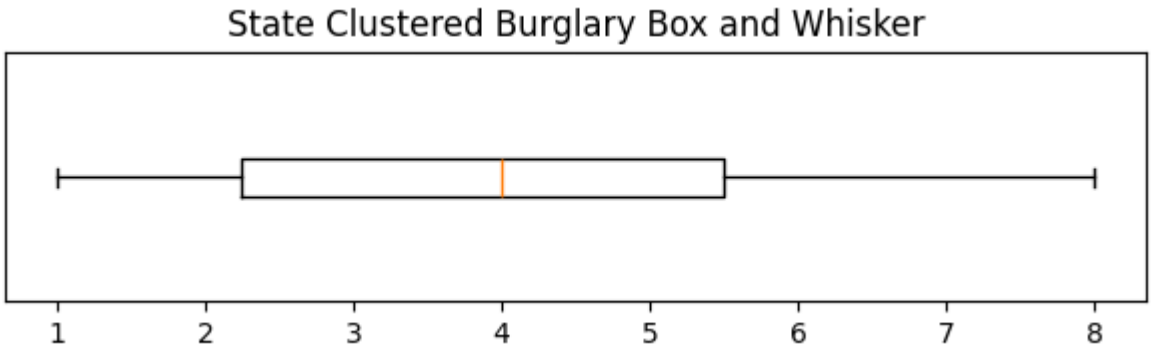


Figure 22: Clustered burglary changes boxplot

Continuing by looking at shoplifting data. The state groupings, changes each month per group, total changes from all groups in each month, and the boxplot are shown as Figures A.12, A.13, A.14, and A.15 in the appendix. The states that ended up being clustered together were very similar to burglary with a few differences, notably with Florida being grouped with other states now rather than its own group. Once again, there were no months that were considered to be outliers according to the boxplot, so we can't say that there's a definitive increase in shoplifting for any particular month.

And finally, pickpocket. Similar to shoplifting, the state groupings, changes each month per group, total changes from all groups in each month, and the boxplot are shown as Figures A.16, A.17, A.18 and A.19 in the appendix. Once again, we mostly see similar state groupings, with Florida still being clustered with other states, while Louisiana stands alone, alongside California and Texas. The CUSUM results show that while this data has very low variance, in terms of the changes within individual groups, the total changes per month show a different picture. However, once again, there is demonstrated increase of crime throughout the year, rather than during holiday months, with November being the notable exception to the overall increase. The boxplot once again confirms that there are no true outliers.

City level - Los Angeles

Our city data is more granular, with data points for each individual day allowing for the ability to detect changes in the dates immediately around the holidays while also retaining the ability to be aggregated up to the month level. Starting with Los Angeles (LA) burglary data, the entire series of data is plotted, along with the detected changes below.

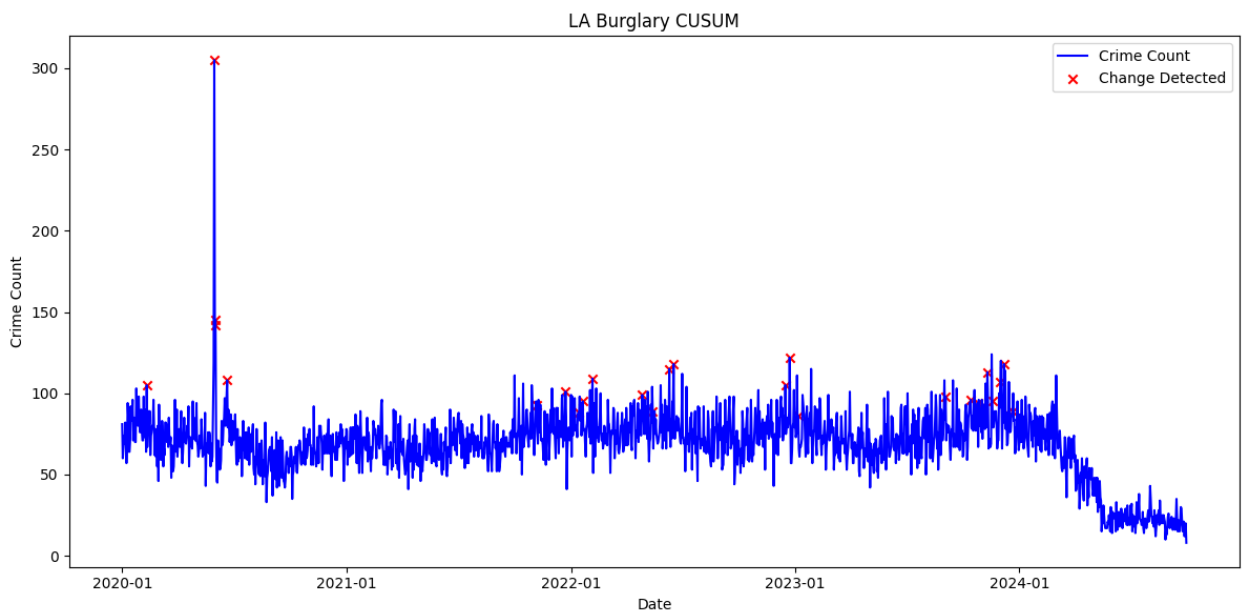


Figure 23: Los Angeles burglary CUSUM

Considering length of the data overall, not that many changes were detected and the only date that does have multiple changes is December 23, but it only has two changes, which is not enough to be considered a true increase in crime. Aggregating the changes to the month level and visualizing the sum as a histogram and boxplot are shown below.

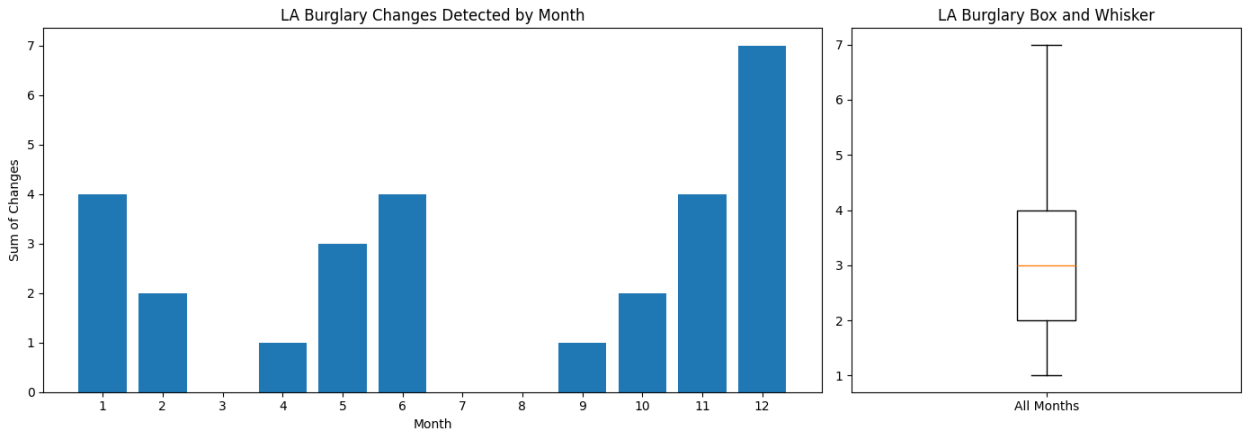


Figure 24: Los Angeles burglary changes per month and boxplot

Once again, while the eye test might suggest a month, in this case December, to be higher than others, the boxplot doesn't consider it as a true outlier.

The aggregated results for LA pickpocket and shoplifting data are shown below and the plots of all the data and changes for each crime type can be found in Figures A.20 and A.21 in the appendix.

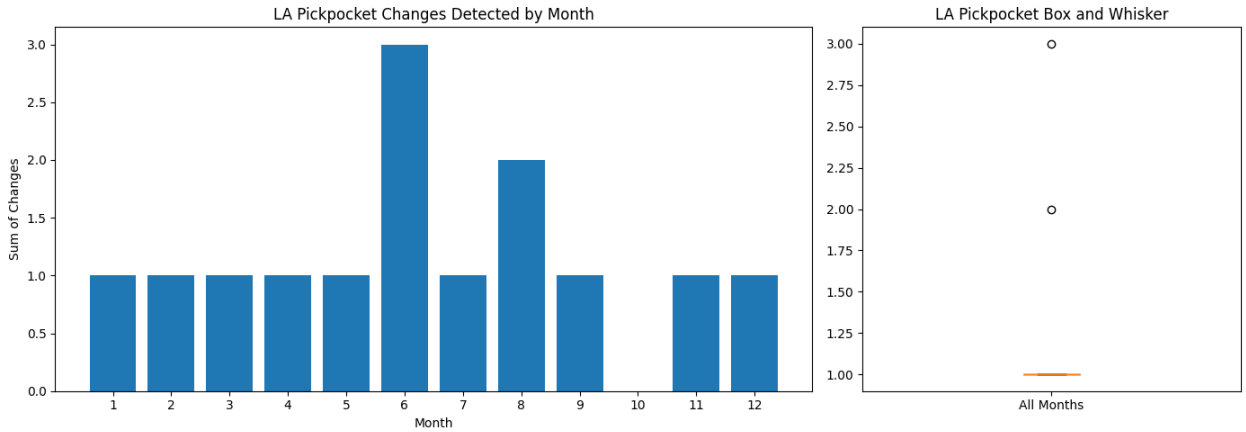


Figure 25: Los Angeles pickpocket changes per month and boxplot

Notably, the pickpocket data does technically have outliers detected in the boxplot, but the month with the most extreme values is June, which we don't consider a holiday month, and it only has three changes, compared to one change per month, excluding August and October.

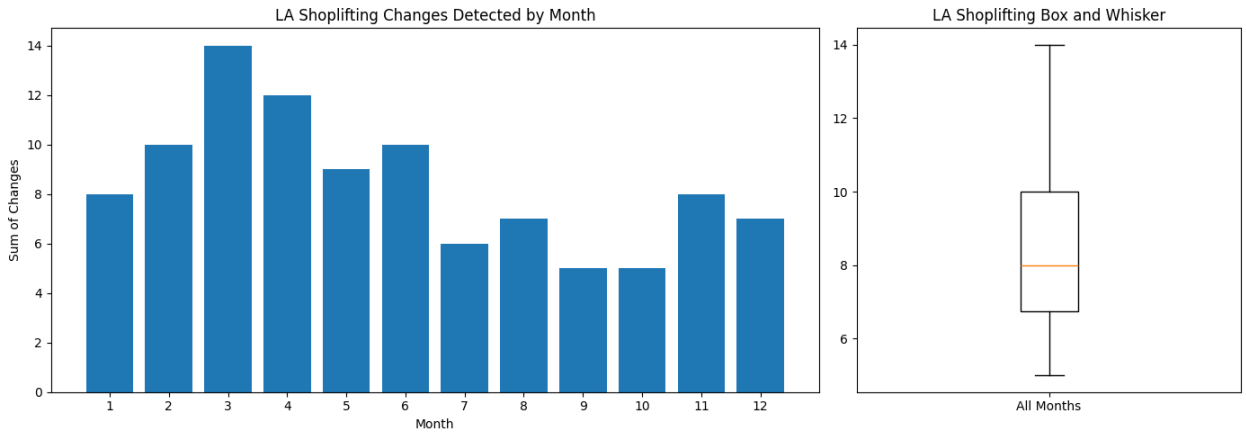


Figure 26: Los Angeles shoplifting changes per month and boxplot

Shoplifting has a lot more occurrences of changes, but relatively an even spread across months, so no outlier is detected.

City level – Boston

Boston aggregated results are shown below for all three types of crime with individual plots for each crime type included as figures A.22, A.23, and A.24 in the appendix.

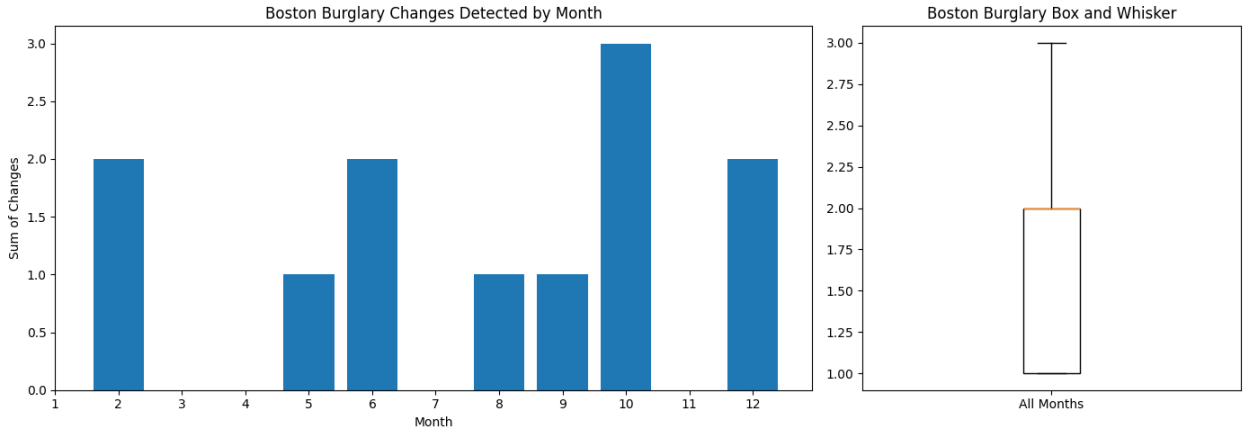


Figure 27: Boston burglary changes per month and boxplot

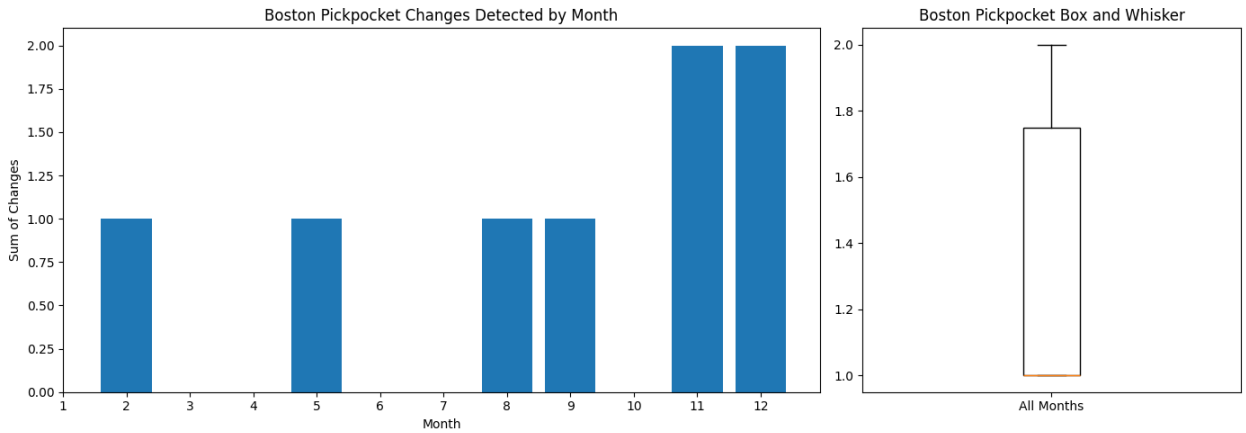


Figure 28: Boston pickpocket changes per month and boxplot

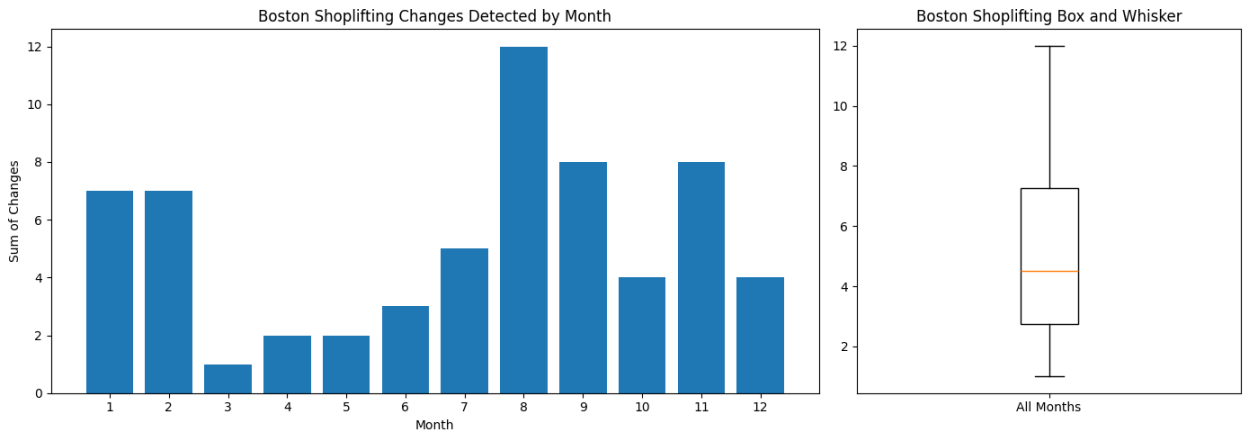


Figure 29: Boston shoplifting changes per month and boxplot

Just like for LA, we do have plenty of instances of change detected by CUSUM, but no month has enough to be considered an outlier, so we still aren't able to support the original hypothesis that crime levels are higher during the holiday months.

Conclusion

This study applied SARIMA models and machine learning techniques (XGBoost) to forecast national-level shoplifting, burglary, and pocket-picking cases, as well as to detect anomalies during holiday months. Our results indicate that shoplifting and pocket-picking cases exhibit an upward trend, while burglary cases display seasonal fluctuations. Model evaluation metrics (RMSE, MAE, and MAPE) suggest that SARIMA performed relatively well for shoplifting, achieving the lowest MAPE (3.50%), indicating higher predictive accuracy. In contrast, burglary forecasts showed the highest RMSE and MAE values, reflecting greater variability, while pocket-picking forecasts had the highest MAPE (6.98%), indicating a larger relative error despite lower absolute errors.

Compared to the SARIMA models, the corresponding XGBoost models performed poorly on the national petty crime data. This may be due to the strong seasonality effects, which SARIMA

models capture more effectively than XGBoost, as the latter does not inherently model seasonality.

Anomaly detection results revealed that December 2023 had significantly lower-than-expected burglary and pocket-picking cases, suggesting that external factors may have contributed to crime reductions during this period. In contrast, shoplifting cases remained within expected ranges, with no extreme anomalies detected. While the SARIMA model captured overall trends well, it tended to underestimate crime spikes, particularly during peak months, highlighting the need for more adaptive forecasting techniques.

Our CUSUM analysis, in combination with using boxplots, was able to act as a confirmation of the results provided by SARIMA and XGBoost. Following the trend set by those models, CUSUM showed us that while changes in crime levels do occur, the boxplots determined that there's no significant difference in the number of changes in each month that supported our initial hypothesis that crime increases during holiday periods. This was true while looking at individual states, the clusters of states formed using K-means, and both cities in focus.

Future research should investigate external factors (e.g., policy changes, economic conditions, law enforcement efforts) that may have contributed to the observed crime reductions in December. Additionally, incorporating Bayesian models could enhance predictive accuracy by dynamically adjusting to recent crime trends. Including exogenous variables, such as law enforcement reports, socioeconomic indicators, and major events may further improve forecasts and enhance anomaly detection. These advancements can support law enforcement agencies and policymakers in developing data-driven strategies for crime prevention and resource allocation.

References

A. Mason, Y. Zhao, H. He, R. Gompelman and S. Mandava, "Online Anomaly Detection of Time Series at Scale," *2019 International Conference on Cyber Situational Awareness, Data Analytics and Assessment (Cyber SA)*, Oxford, UK, 2019, pp. 1-8, doi: 10.1109/CyberSA.2019.8899398

Fang, T., & Lahdelma, R. (2016). Evaluation of a multiple linear regression model and SARIMA model in forecasting heat demand for district heating system. *Applied Energy*, *179*, 544-552. https://doi.org/10.1016/j.apenergy.2016.06.133

Isbister, T. (2015). Anomaly detection on social media using ARIMA models.

Manuca, R., & Savit, R. (1996). Stationarity and nonstationarity in time series analysis. *Physica D: Nonlinear Phenomena*, *99*(2-3), 134-161.

Pena, E.H.M., de Assis, M. V. O., and Proença, M. L. "Anomaly Detection Using Forecasting Methods ARIMA and HWDS," *2013 32nd International Conference of the Chilean Computer Science Society (SCCC)*, Temuco, Chile, 2013, pp. 63-66, doi: 10.1109/SCCC.2013.18.

Petrică, Andreea-Cristina, Stelian Stancu, and Virgil Ghițulescu. "Stationarity–The central concept in time series analysis." *International Journal of Emerging Research in Management and Technology* 6, no. 1 (2017): 6-16.

Ramraj, S., Uzir, N., Sunil, R., & Banerjee, S. (2016). Experimenting XGBoost algorithm for prediction and classification of different datasets. *International Journal of Control Theory and Applications*, *9*(40), 651-662

Sen, N., Temur, L. O., & Atilla, D. C. (2024). Yellow Fever Vaccine Demand Forecasting with ARIMA, SARIMA, Linear Regression and XGBoost. *IEEE Access*.

Thejovathi, M., ChandraSekharaRao, M. V. P., Priyadharsini, E. J., Siddi, S., Karthik, B., & Abbas, S. H. (2024). Optimizing Product Demand Forecasting with Hybrid Machine Learning and Time Series Models: A Comparative Analysis of XGBoost and SARIMA. *EJ and Siddi, Someshwar and Karthik, B. and Abbas, Syed Hauider, Optimizing Product Demand Forecasting with Hybrid Machine Learning and Time Series Models: A Comparative Analysis of XGBoost and SARIMA (November 15, 2024)*.

Wu, W., Guo, M., Wang, S., & Han, J. (2025). Time series anomaly detection hybrid model based on SARIMA and LSTM. *Journal of Computing and Electronic Information Management*, *16*(1), 63-69.

Appendix

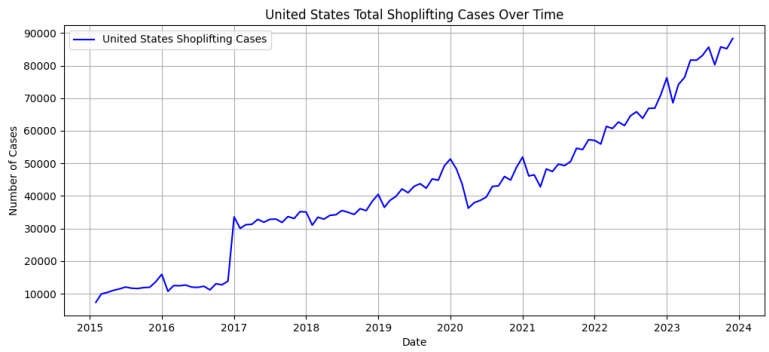


Figure A.1: National shoplifting data from 2015 to 2024

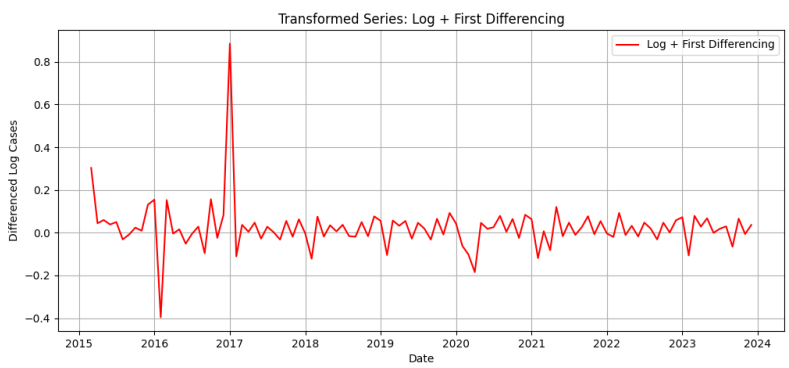


Figure A.2: Log -transformed first differenced shoplifting data from 2015 to 2024

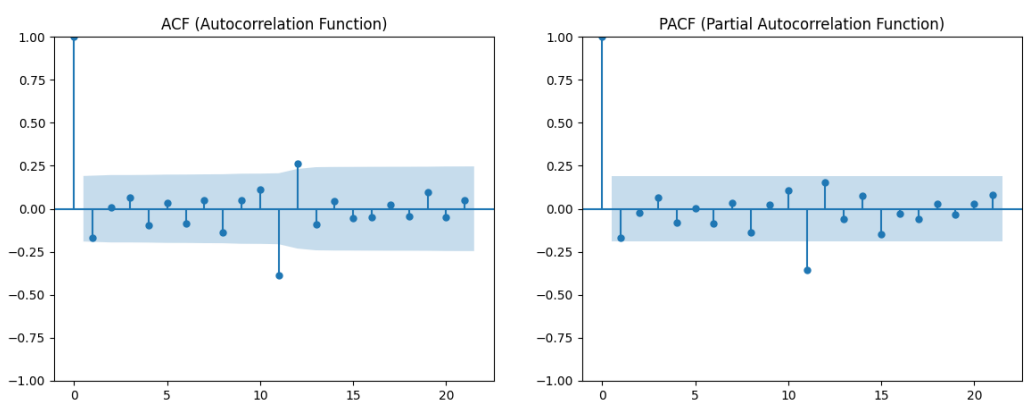


Figure A.3: ACF and PACF plots

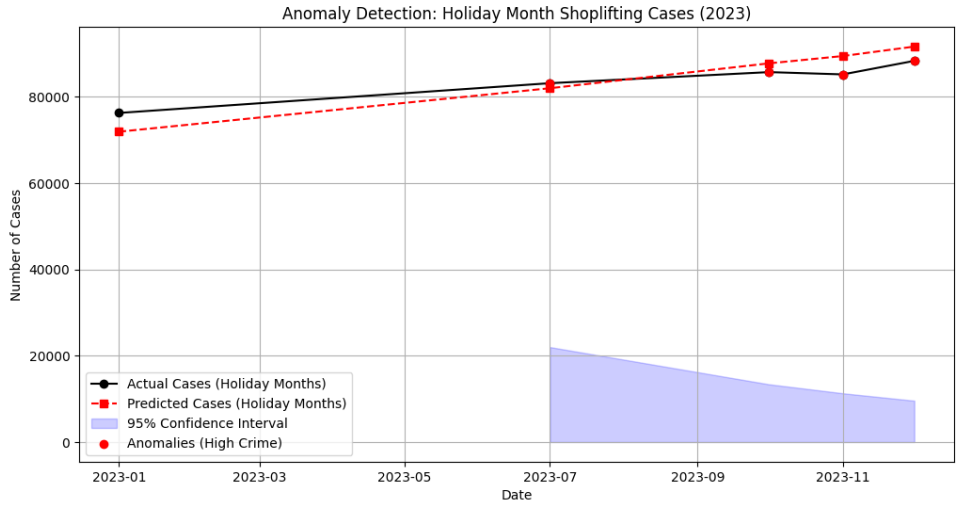


Figure A.4: Anomaly detection for shoplifting using 2023 data

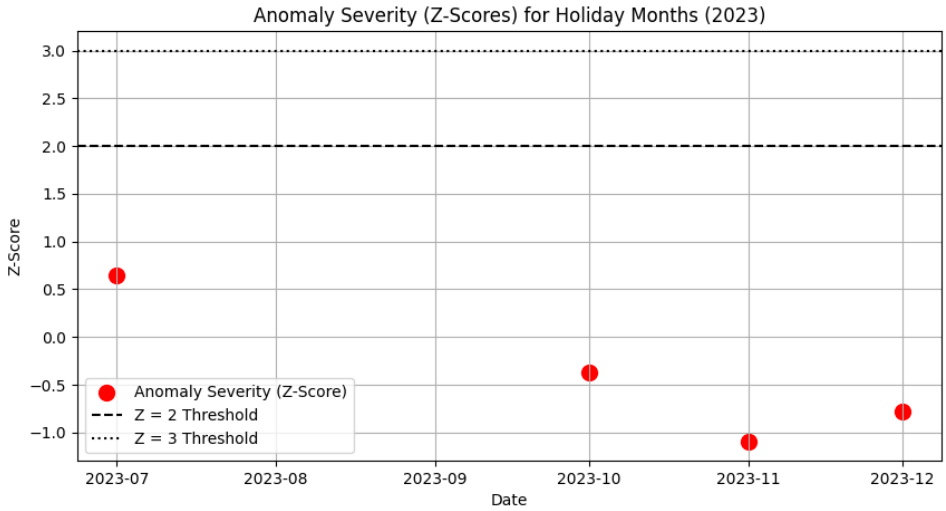


Figure A.5: Anomaly detection for shoplifting using z-score method for 2023

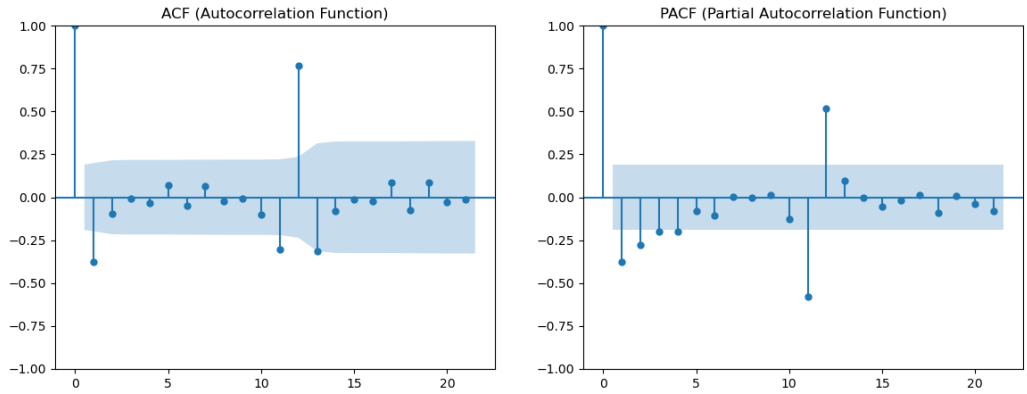


Figure A.6: ACF and PACF plots for national level burglary data

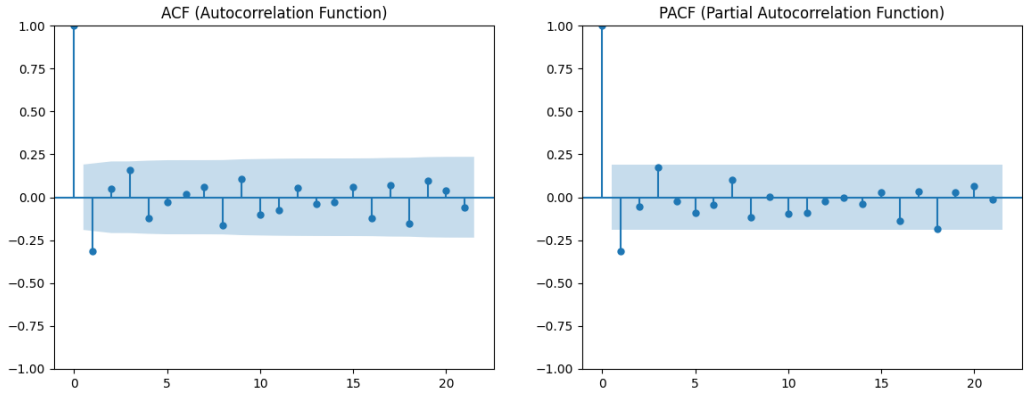


Figure A.7: ACF and PACF plots for national level pocket-picking data

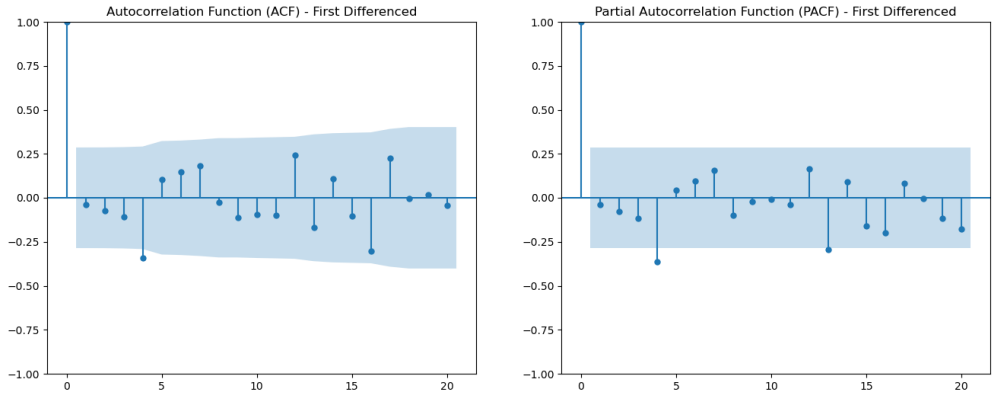


Figure A.8: ACF and PACF plots for LA burglary cases data

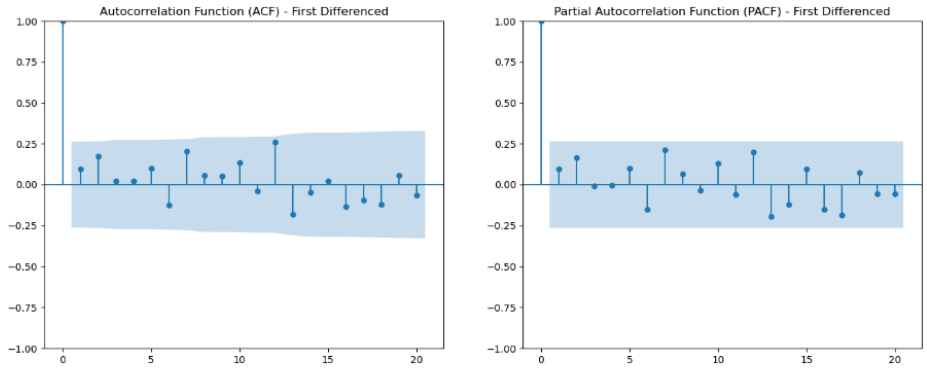


Figure A.9: ACF and PACF plots for LA shoplifting cases data

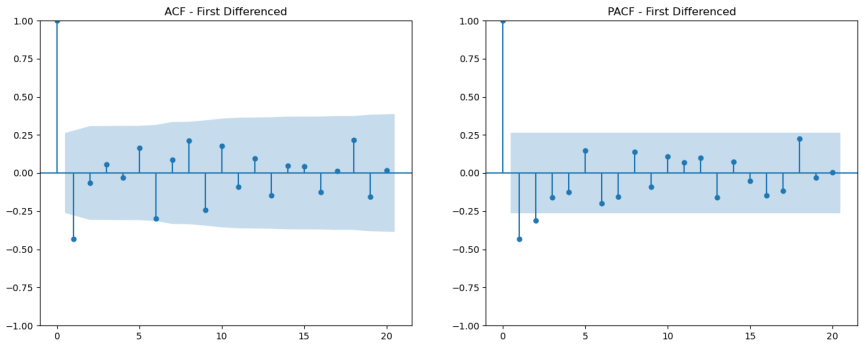


Figure A.10: ACF and PACF plots for LA pocket-picking cases data

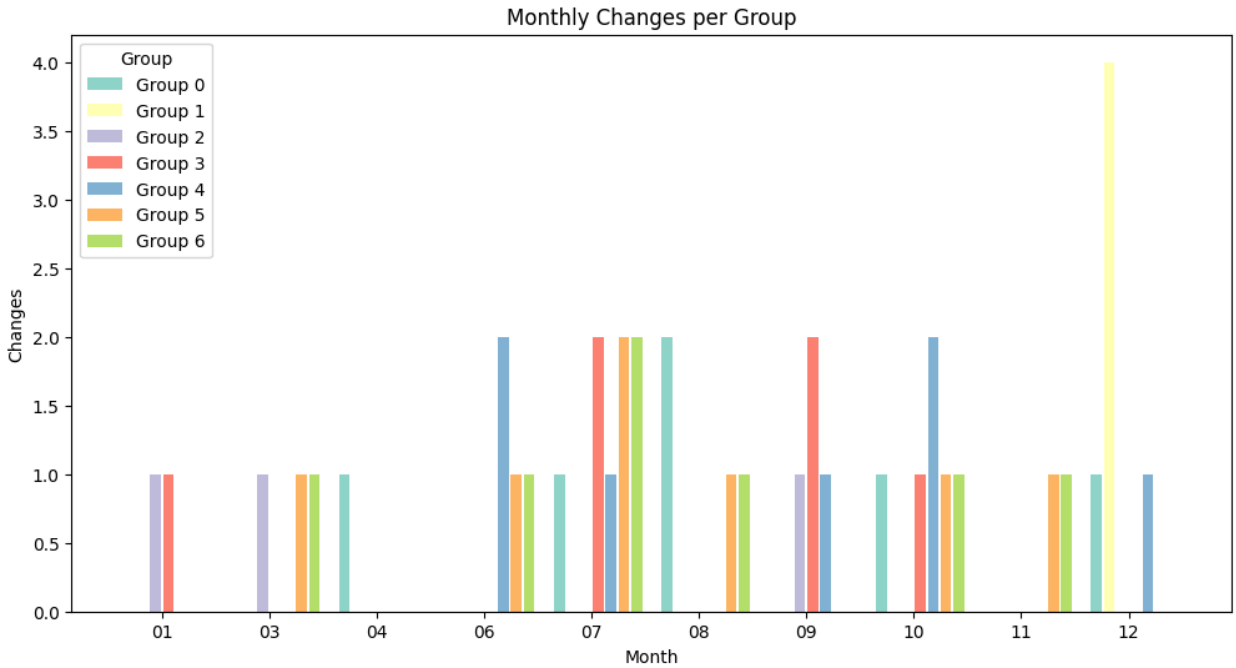


Figure A.11: Burglary month of detected changes from clustered states

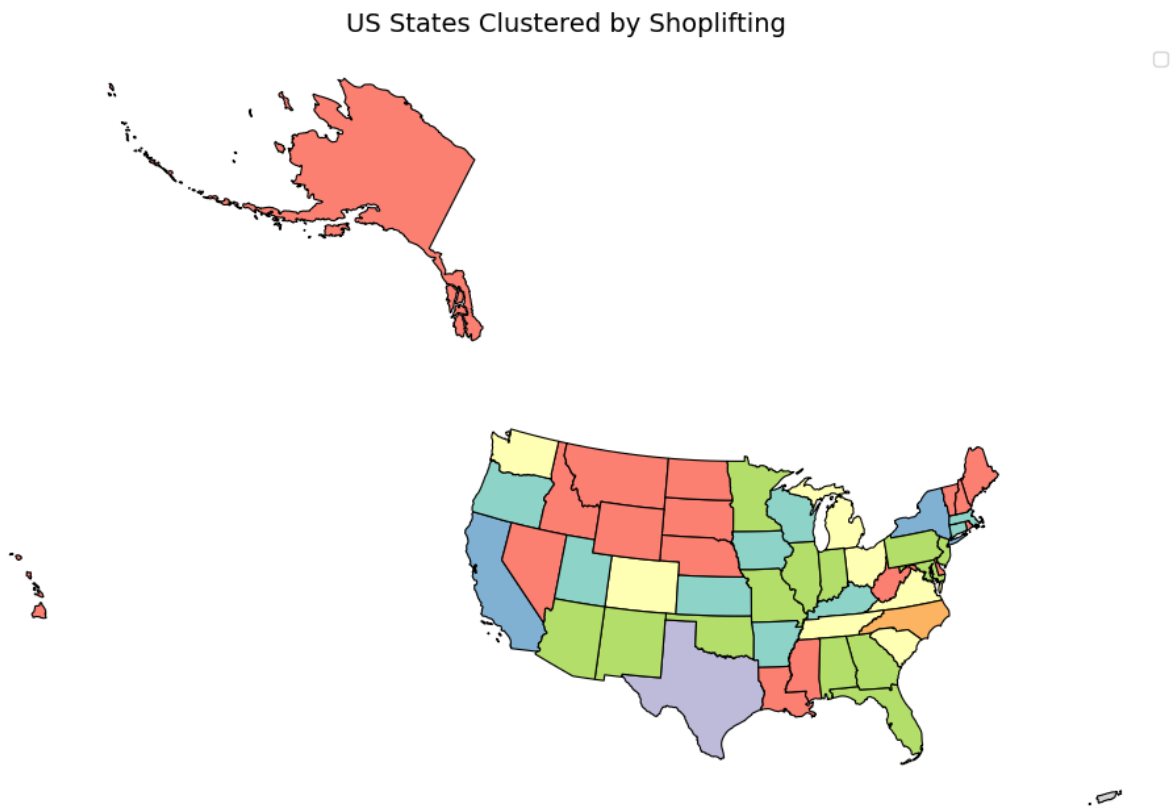


Figure A.12: Shoplifting state clustering results

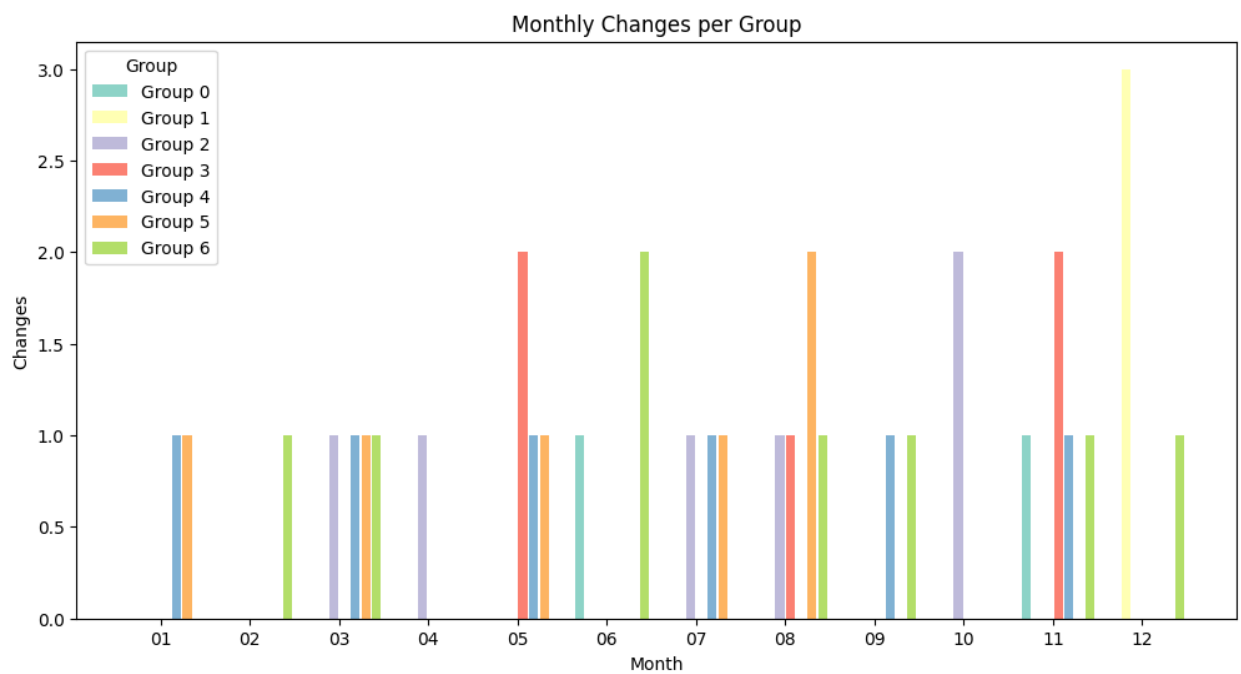


Figure A.13: Shoplifting month of detected changes from clustered states

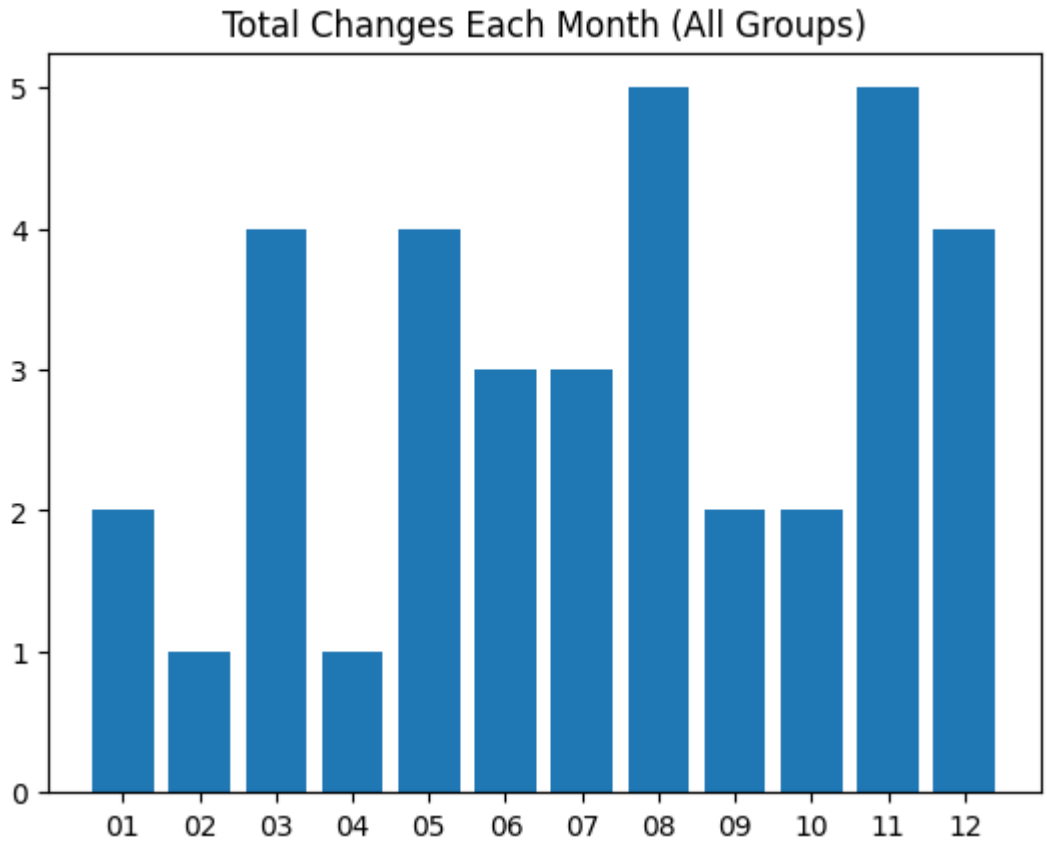


Figure A.14: Shoplifting total changes detected each month across all clusters

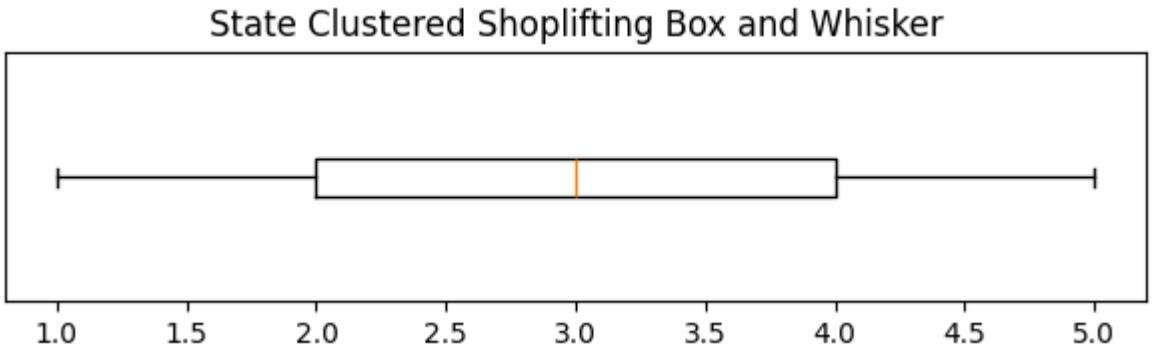


Figure A.15: Shoplifting boxplot for changes detected each month

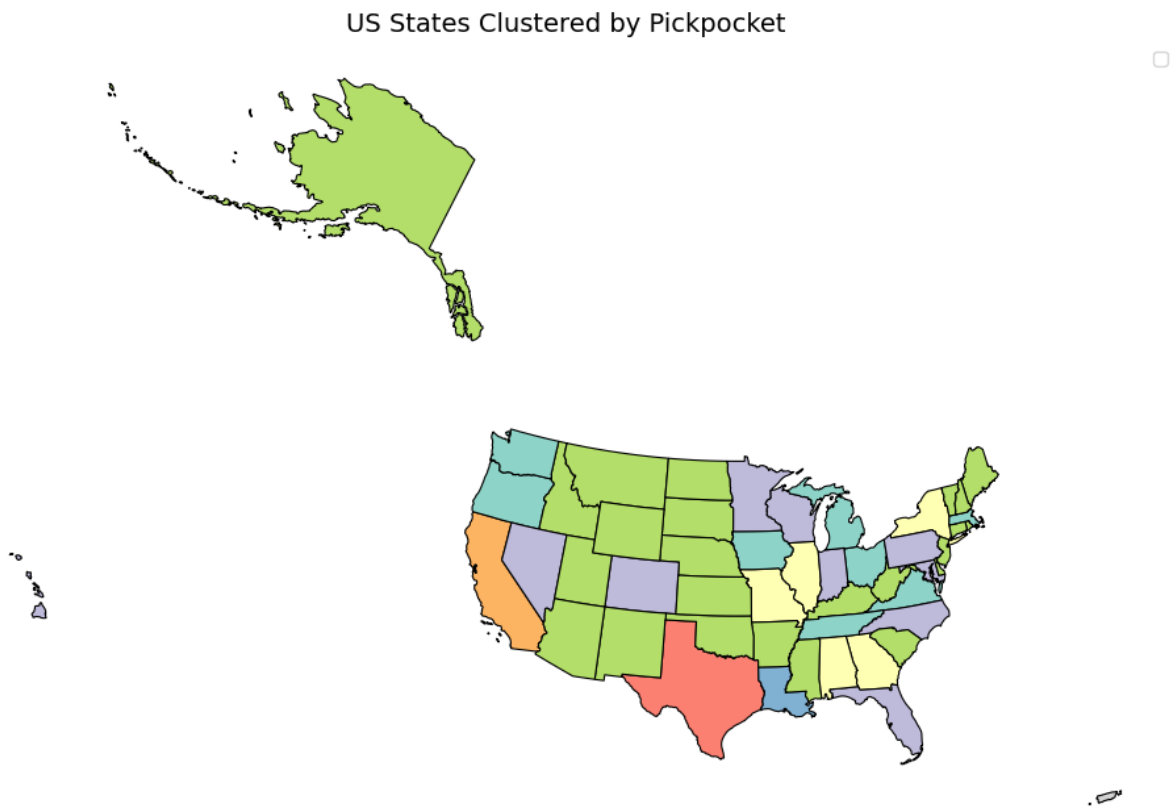


Figure A.16: Pickpocket state clustering results

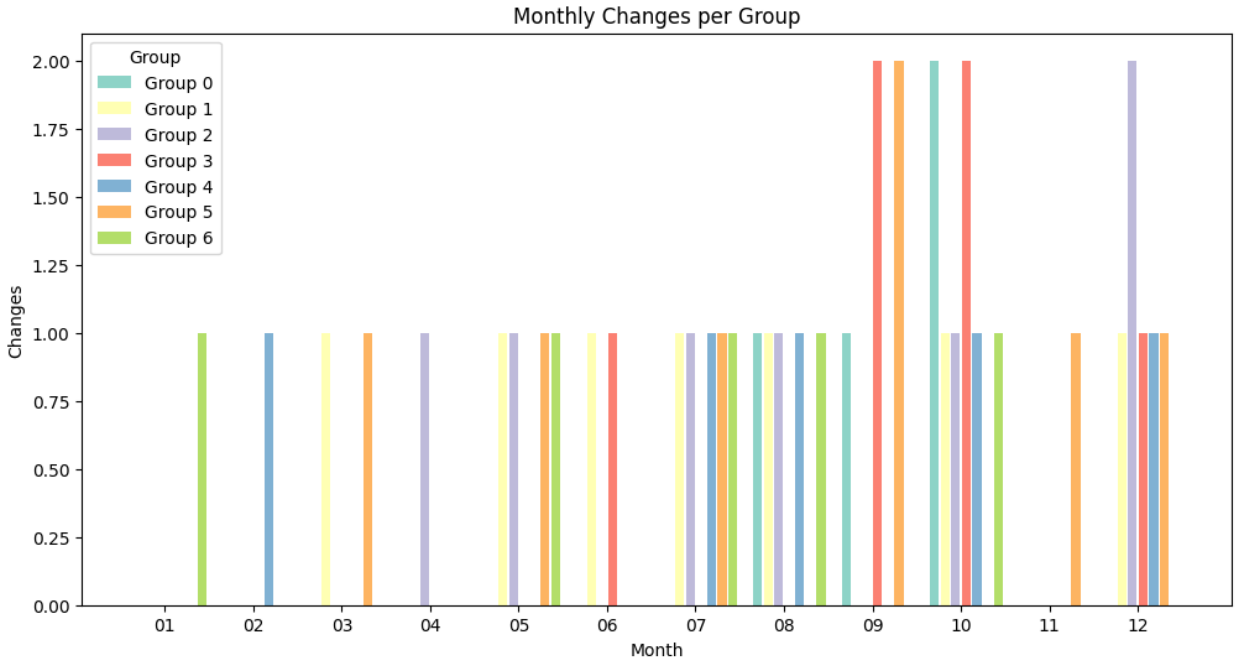


Figure A.17: Pickpocket month of detected changes from clustered states

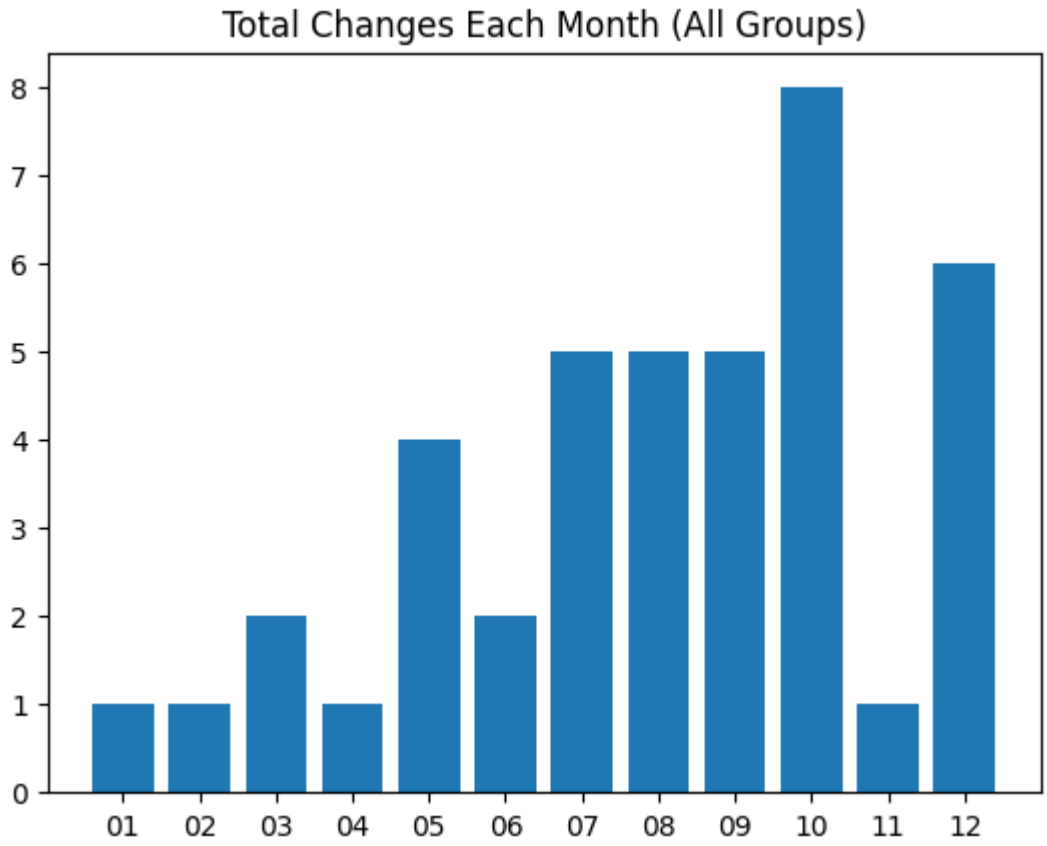


Figure A.18: Pickpocket total changes detected each month across all clusters

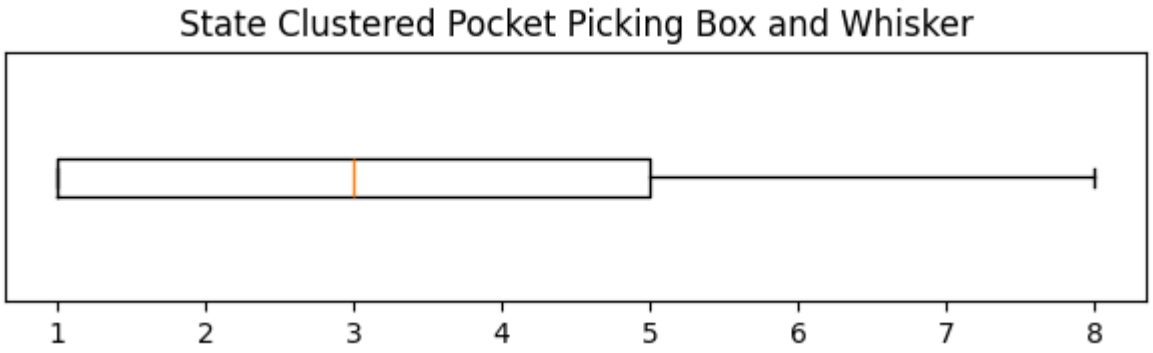
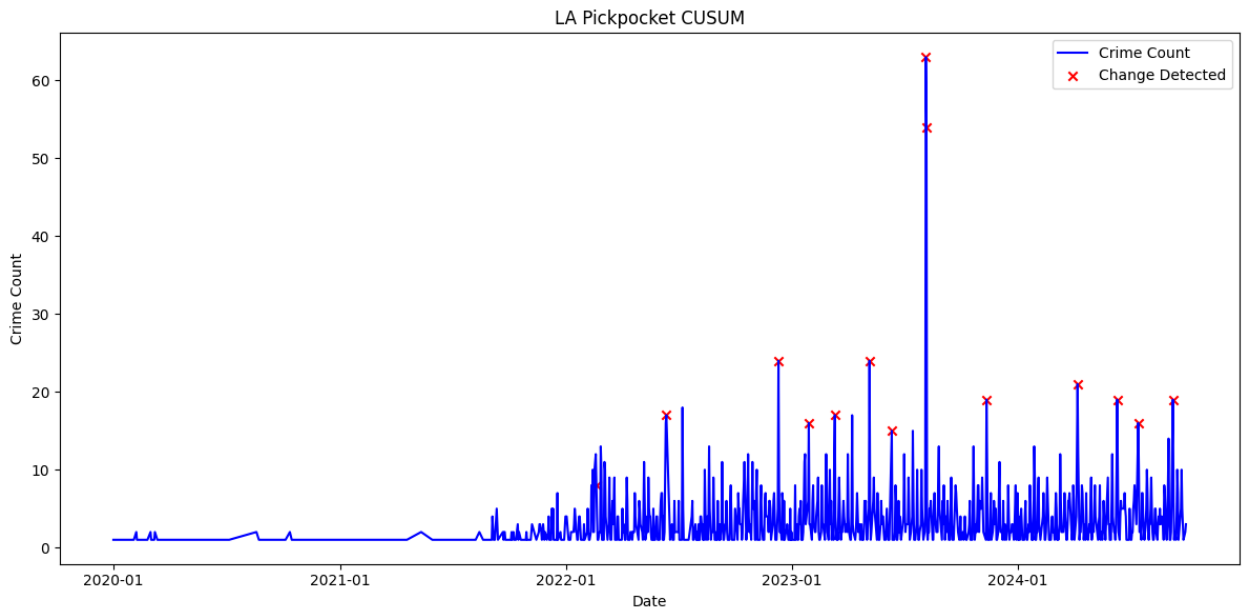
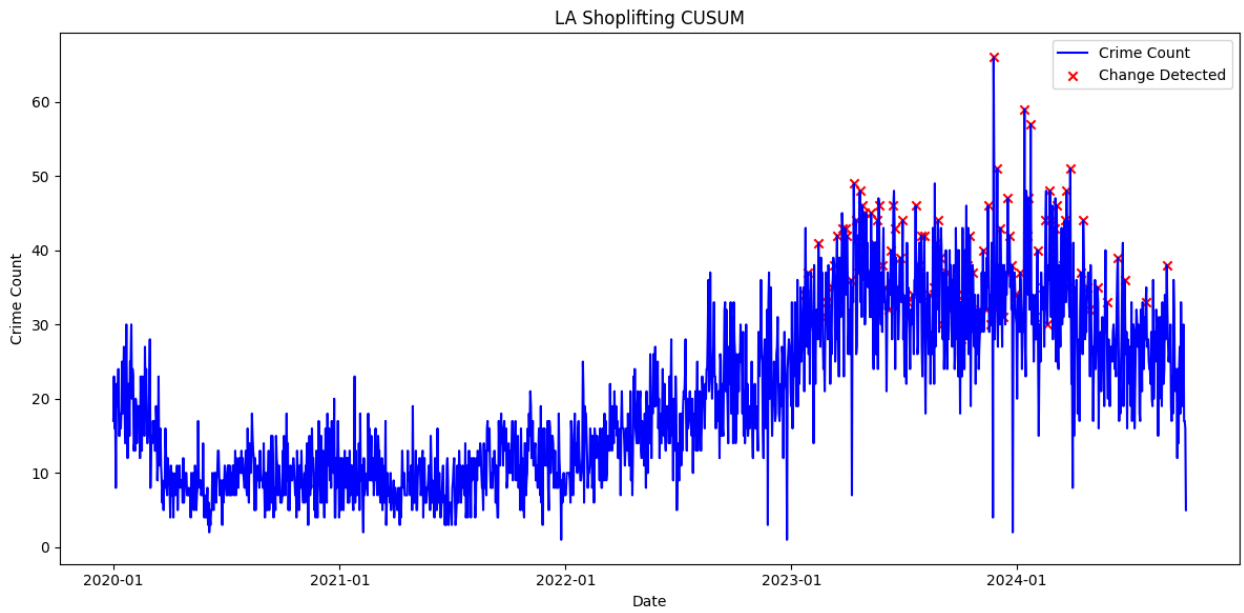


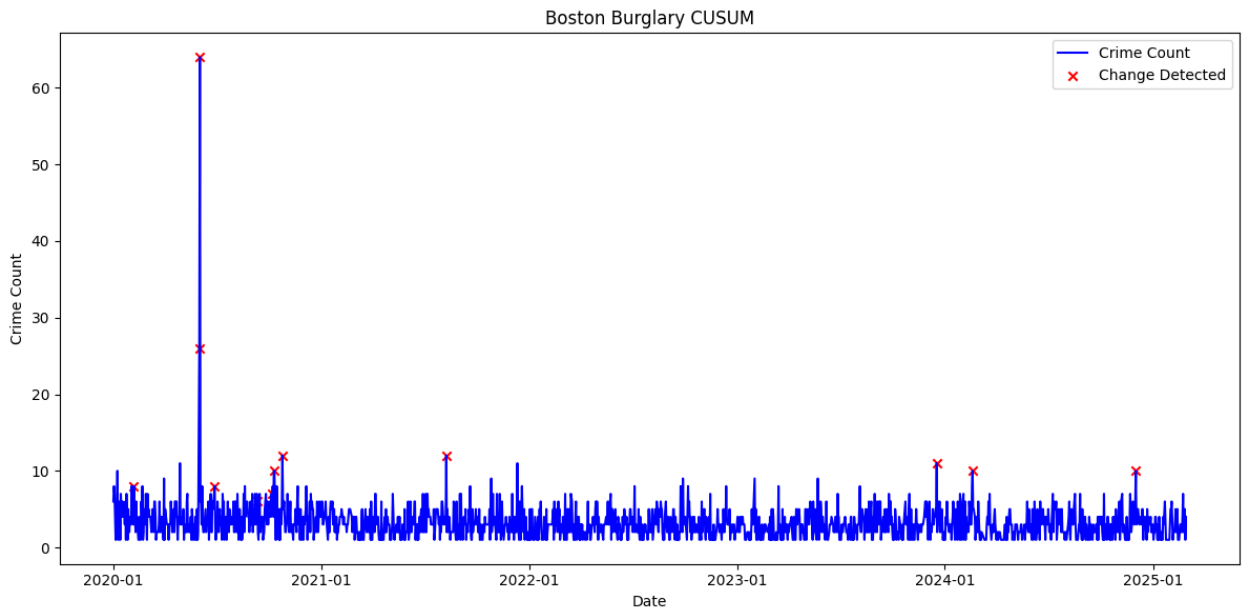
Figure A.19: Pickpocket boxplot for changes detected each month



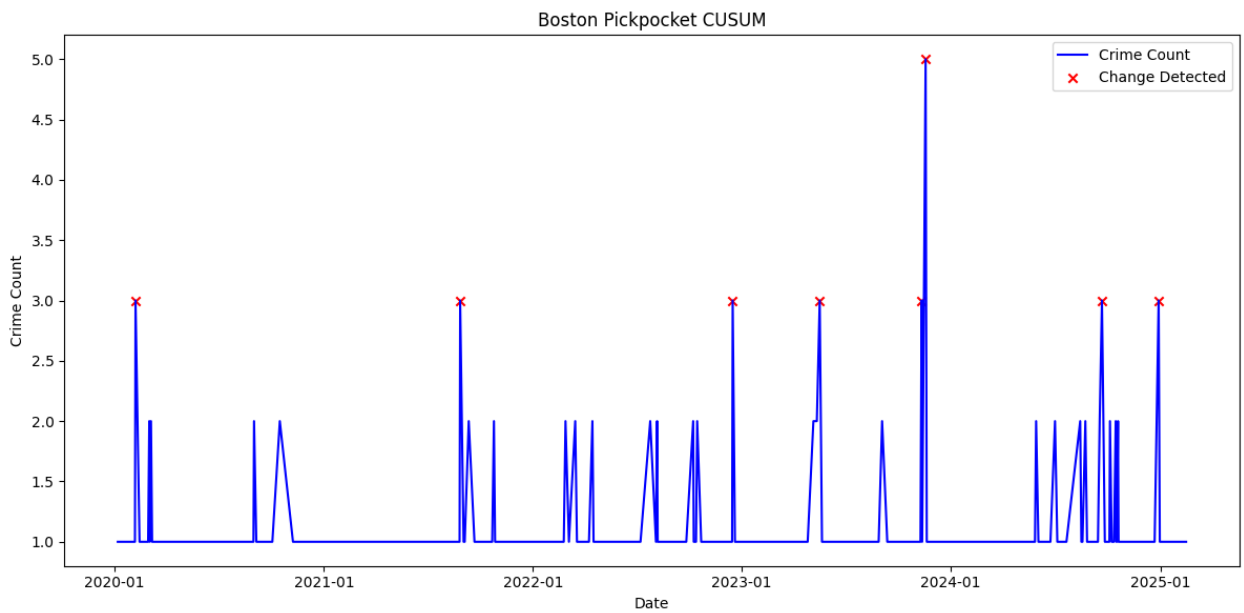
A.20: LA Pickpocket CUSUM analysis



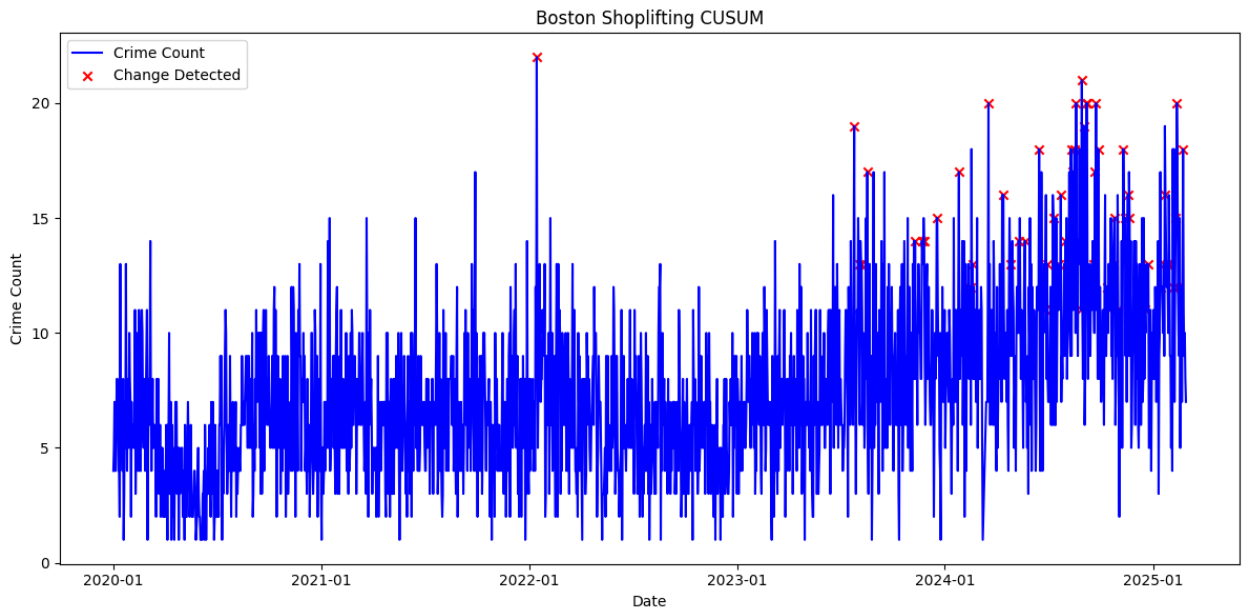
A.21: LA Shoplifting CUSUM analysis



A.22: Boston Burglary CUSUM analysis



A.23: Boston Pickpocket CUSUM analysis



A.24: Boston Shoplifting CUSUM analysis